

Section III

Dry Cleaning Processes

Chapter 6

Gas-Phase Wafer Cleaning Technology

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6.1 INTRODUCTION AND OVERVIEW

In this chapter, cleaning processes that expose the wafer surface to various gas-phase environments will be discussed. Typical 14-nm node technology logic devices require over 158 cleaning steps during manufacturing, accounting for 16% of the total process steps, one of the most repeated processes in integrated circuit (IC) manufacturing. Although cleaning, in general, is a pervasive requirement, each of the cleaning steps carries its own set of contaminant removal and selectivity requirements and thus may require different approaches and technologies. Although liquid-phase cleaning will continue to be used for the majority of the cleaning steps, gas-phase cleaning has found application in meeting the selectivity needs of specific cleaning steps. Gas-phase cleaning technology spans a wide variety of techniques. This handbook attempts to provide broad coverage of these techniques; plasma, cryogenic aerosol, and supercritical fluid cleaning techniques will be discussed in detail in Chapters 7 and 8. Readers may have seen the term “vapor-phase” used to refer to processes carried out at or near atmospheric pressure with the gaseous reactants near the saturation point. In this chapter, the term “gas-phase” will include “vapor phase” processes and processes carried out under vacuum conditions.

The use of ultraviolet (UV) irradiation is an important addition to many gas-phase cleaning processes. The capability of UV light to decompose organic molecules has been known for a long time. Additionally, UV cleaning of surfaces has been explored since the mid-1970s [1–4]. Since 1976, use of the UV/O₃ (ozone)-cleaning method has grown steadily. UV/O₃ cleaners are now available commercially from several manufacturers. As shown in this chapter, UV is also used to activate other gas-phase chemistries, such as Cl₂.

6.1.1 Limitations of Wet Cleaning

A general overview of wafer cleaning is covered in Chapter 1. Wet wafer cleaning will likely remain the dominant technique for the removal of contaminants from Si and SiGe surfaces in IC manufacturing. Several factors, however, are driving a continued interest in the alternative methods of gas-phase wafer cleaning [5,6].

First, wet wafer cleaning operations present challenges for process integration. Clustering of surface preparation chambers with reduced pressure deposition and thermal and etching operations will become easier as hardware for gas-phase removal of trace contaminants become available [7]. Second, problems are anticipated with wet processing, in particular aqueous processing of high-aspect ratio structures. The complications stem not from the difficulty of getting liquid into small openings, as this problem can be dealt with by using surfactants, but from the difficulty of getting it out [8]. Once forced into deep trenches and contacts, cleaning liquids can only be removed by evaporation, which can leave contaminants behind. Furthermore, the capillary forces that are present during the drying of these structures might result in their collapse with negative yield loss [9]. Third, to continue scaling, CMOS (complementary metal oxide semiconductor) technology is moving to new architectures such as FinFET (fin field effect transistors) and nanowire field effect transistors. These architectures are nonplanar and have spurred the development of new cleaning processes. One example is isotropic etching processes that remove a controlled amount of material—perhaps as little as one monolayer—from the channel after it is formed. Fourth, large improvements in performance are possible with nonplanar architectures using semiconductors that have higher charge mobilities than silicon (Si). Silicon germanium (SiGe) alloys are especially promising as a channel material because the carrier mobility can be tuned by changing the percentage of Ge. Integrating SiGe into CMOS technology requires an understanding of the similarities and differences between the way Si and Ge react. For instance, Ge oxide readily etches and dissolves in water, whereas Si oxide has nearly a zero etching rate in water [10,11].

Next, there is the difficulty of controlling particles during wafer treatments in liquid chemicals and rinse—dry cycles. Use of “all dry” cleaning is expected to reduce this problem substantially, because particles are easier to control in gases than in liquids. Furthermore, the replacement of liquid cleaning agents with their gaseous counterparts is expected to result in a decrease in the overall level of surface contamination because at least part of the contamination originates from the wet cleaning environment itself [12]. Metallic contaminants, for instance, can be plated on Si surfaces from HF solutions [13] and deposited from NH_4OH and other agents used in wet cleaning, such as H_2O_2 [14–16]. However, organic contaminants originate in part from the containers and utensils used in wet-chemical operations [17]. Finally, the cost of high-purity

chemicals and DIW (deionized water), problems with waste disposal, and safety issues are additional reasons for the continuing interest in gas-phase and other dry wafer cleaning technology.

Attempts have been made to overcome some of these wet processing problems. These approaches include spray cleaning [18], displacement processing [19], jet aerosol cleaning [20], megasonic-based cleaning [21], chemical recirculation [22], and others [23]. Chapter 4 discusses the limitations of wet cleaning.

6.1.2 Drivers for Gas-Phase Cleaning and Surface Conditioning

Surface preparation or cleaning accounts for approximately 16% of all processing steps used to build a device. Cleaning isolates the primary fabrication steps from one another to achieve a high yield of functional devices. For example, deposition, etching, and ion implantation processes alter the surface, leaving it in an unacceptable condition for the subsequent step in a device flow. The goal of cleaning is to optimize the chemical, material, and electrical properties of a surface for the subsequent processing step.

In general, it is not expected that gas-phase will replace liquid-phase cleaning applications, in which the latter is successfully used. Instead, gas-phase cleaning should be considered for those applications in which wet cleaning, as a result of its inherent limitations, cannot be used. First and foremost, this concerns in situ cleaning operations carried out within integrated processor systems. An example of an early application of gas-phase cleaning in cluster tool manufacturing is shown in Fig. 6.1-1, in which an atmospheric

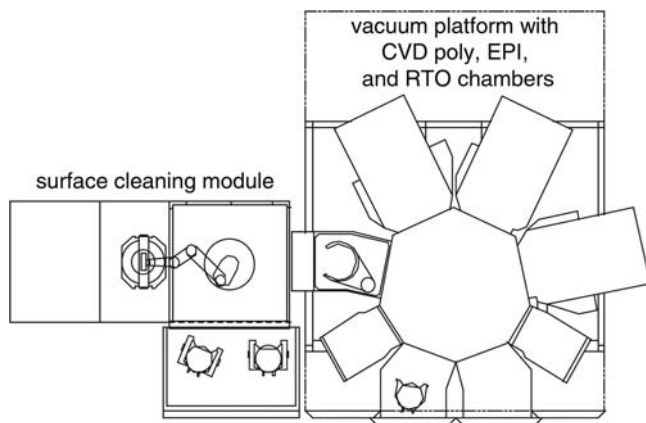


FIGURE 6.1-1 Cluster tool used for early studies of integrated cleaning for gate stack deposition and epitaxial Si deposition. An atmospheric pressure gaseous $\text{HF}/\text{H}_2\text{O}$ process is used in the surface cleaning module and wafers are transferred through a loadlock into the vacuum platform for gate dielectric growth and polySi gate deposition or epitaxial Si deposition [24]. (Used with permission from Semiconductor International.)

pressure gaseous $\text{HF}/\text{H}_2\text{O}$ process is clustered onto a vacuum platform [24]. This cluster system is used for gate stack deposition and epitaxial Si deposition. Other likely applications for a cluster tool with integrated cleaning include precontact, preepitaxial, SOI (Si on insulator), SiGe, and other cleans, such as those applied before polySi emitter formation in bipolar technology [7]. Overall, it can be expected that gas-phase cleaning modules will be used predominantly in conjunction with cluster tools. Refinements of gas-phase wafer cleaning methods are also leading to the use of gas-phase cleaning reactors in the stand-alone mode.

In further defining the potential role of gas-phase cleaning in IC processing, the ability of both gas-phase and liquid-phase cleans to remove specific surface contaminants should be considered. Clearly, the most challenging contaminants are metals, including transition and heavy metals, and alkali and alkali earth ions. Gas-phase cleaning may not be able to remove large amounts of these contaminants without substantial etching of Si and/or roughening the surface. Therefore, gas-phase cleaning processes are not expected to be as effective as liquid-phase cleaning processed on surfaces grossly contaminated with metals.

Considering the nature of various cleaning modes, it may be appropriate to separate them into: (1) treatments that are expected to remove large amounts of surface contaminants and (2) treatments that, besides removing trace contaminants, change the chemical state of the surface. An example is the removal of native/chemical oxide or surface passivation through oxidation or hydrogenation. In this latter case, the term “surface conditioning” may be more appropriate than “surface cleaning.” The nature of such surface conditioning processes will have to be closely related to the requirements of the subsequent processing step.

Following this reasoning, the role of processes currently referred to as gas-phase cleaning may be limited in the future primarily to in situ surface conditioning, whereas liquid-phase cleaning will continue to be carried out as ex situ surface cleaning operations. If the current cleaning trends continue, the emphasis will gradually shift toward in situ processing with improved overall process cleanliness with less emphasis placed on surface cleaning in the traditional meaning of this term. To become technically viable, any gas-phase cleaning method will have to meet certain requirements, which is briefly summarized as follows:

1. Gas-phase cleaning methods must display experimentally proven capability to remove contaminants comparable with the performance of liquid-phase cleans. This requirement applies in particular to the removal of heavy, transition, and alkali metals that, among other contaminants of concern, are the most difficult to volatilize. No general-purpose gas-phase cleaning will be considered effective unless it displays experimentally proven ability to remove trace metals.

2. Gas-phase cleans must not roughen the semiconductor surface nor generate defects in the oxide. This issue is particularly relevant to cleaning processes that are based on aggressive, plasma-generated chemistries and/or very short wavelength electromagnetic radiation. Consequently, a careful assessment of gas-phase cleaning environment interactions with surface features must be carried out.
3. Any gas-phase cleaning method must have adequate throughput, ideally high enough to make the cleaning steps compatible with batch furnace processing. Further discussion in this chapter will reveal that gas-phase cleaning is typically a single-wafer process. Therefore, throughput is an important issue not only for integrating gas-phase cleans with batch processes but also for implementing cluster tool processing.
4. Gas-phase cleaning methods have to be effective at fairly low temperatures for two reasons. First, they have to meet low-thermal budget processing. Second, elevated temperature at the early stage of gas-phase cleaning process may have an adverse effect on surface contaminants, which, instead of being removed, can diffuse and become permanently embedded in the semiconductor surface or diffuse into the oxide.
5. Gas-phase cleaning modules must be designed to assure compatibility with cluster tool processing and integrated process automation. As mentioned earlier, it is expected, at least for the time being, that gas-phase cleaning modules will be primarily used as a part of an integrated systems rather than as stand-alone units.

6.1.3 Gas-Phase Cleaning Processes

As early as 1960, when epitaxial technology was developed, it was found that a gas-phase in situ HCl (hydrochloric acid) predeposition etch improved the epitaxial film quality [25–28]. Soon, process engineers were experimenting with various gases, such as chlorine gas (Cl_2) [29], HF (hydrofluoric acid), and HI (hydroiodic acid) [30], and other halide compounds [31,32]. Because of the high temperatures required, it was not long before some sort of excitation was employed, such as reactive ions, UV light, plasma, electron cyclotron resonance, and RF (radio frequency) sputter or microwave radiation, to lower the process temperature. Although these processes have demonstrated some value in wafer cleaning, it became apparent that various problems are associated with them [5,6,33]. The main difficulty has to do with adverse radiation effects on the device structures [34–36]. In addition, various contamination problems are observed, both from bombardment of materials in the systems and as a result of the formation of undesirable reaction byproducts from the complex chemistries involved. Some of these byproducts, such as polymers, are extremely difficult to remove from wafer surfaces.

The goals of semiconductor wafer cleaning process are to remove the following from the surface: (1) organic contaminants; (2) metallic contaminants

TABLE 6.1-1 Contaminants and Their Link With the Gas-Phase Process Used to Remove Them From the Surface

Contaminate	Gross Organics	Light Organics	Metals	Native and Chemical Oxide
Vapor		HCl and HF/H ₂ O	HCl and HF/H ₂ O	HF/H ₂ O, HF/CH ₃ OH
Photo	UV/O ₃	UV/O ₃	UV/Cl ₂	UV and HF/H ₂ O

including transition, heavy, and alkali metals; (3) particulate contamination; and (4) native or chemical oxides spontaneously formed on the surface during air exposure or during cleaning. A detailed description of wafer cleaning requirements is outlined in Chapter 2. [Table 6.1-1](#) depicts these contaminants that gas-phase reactions target and links them with the gas-phase reactions that can be used to remove them from a surface.

For thermally enhanced cleaning, externally generated thermal energy delivered to the wafer breaks down gaseous molecules. The atomic species released in this process subsequently react with metallic and other impurities on the wafer surface before they can diffuse into the surface layers. Typically, moderately high temperatures are needed to promote reaction. If combined with low pressure, thermal treatments can, in some cases, result in the removal of surface contaminants by sublimation. An example of such a process is the thermal decomposition of SiO₂ (silicon dioxide) throughout the formation of volatile SiO (silicon oxide) [37]. One important consideration in thermally enhanced cleaning is, as indicated earlier, to trap, or preferably, to volatilize metal contaminants at the surface before they are driven into the Si or the oxide. Because both the cleaning process and diffusion are thermally activated, the Si wafer should not be subjected to elevated temperatures without a gas present.

The cleaning mode defined in [Table 6.1-1](#) as “vapor” reflects an effort to replace chemistries used in liquid-phase cleaning with their gas-phase analogs. Instead of using water-based solutions of given chemicals, H₂O (water) vapor is added to the gas stream. A typical example of this type of process is a vapor-phase HF etching of native/chemical SiO_x. In some cases, moisture can form on the surface as a product of the cleaning reaction, although the original reactant gases are anhydrous.

Also listed in [Table 6.1-1](#) is photochemically enhanced cleaning. This approach is gaining in popularity because of its effectiveness and relatively simple equipment. Reactant species are generated through the photolysis of gaseous compounds. Radiation in the UV wavelength range is commonly used, as it corresponds to a wide spectrum of bond energies in compounds of interest used for Si processing. Thus, as indicated in [Table 6.1-1](#), a variety of cleaning chemistries can be coupled with UV-enhanced processes.

Plasma is still another source of energy with a potentially wide range of applications in dry wafer cleaning. Plasma-based cleaning processes are discussed in Chapter 7.

6.1.4 Advantages of Gas-Phase Wafer Cleaning

During the improvement of wafer cleaning processes over a number of years, distinct advantages of gas-phase cleaning have been demonstrated. Some of these are indicated here.

Reduced Contamination: One of the problems associated with conventional aqueous cleaning processes has been the redepositing of contaminants back onto the wafers. This has been observed both in the cleaning solution and at the DIW rinsing and drying steps. In the case of gas-phase cleaning processes, self-contamination has not been observed.

Improved Process Uniformity: Typically, as smaller and controlled amounts of reactants are used in gas-phase etching, better control is achieved for both the specific reaction involved and the overall process. For instance, gas-phase cleaning renders atomic layer etching achievable for a variety of applications [38].

Reduced Chemical Usage and Disposal: An important consideration in today's manufacturing processes is that of environmental effects. Because gas-phase cleaning typically uses less than 1/100 of a given chemical than does aqueous cleaning, considerable reduction in the environmental footprint of a process is possible. An example of how gas-phase cleaning can replace wet cleaning, rinsing, and drying steps in a MOS gate oxidation wafer processing sequence is shown in Fig. 6.1-2 [39].

Versatile Process Variables: Gas-phase processing allows much more versatility for process variables than do wet or aqueous systems. For instance, it is much easier to vary temperature and pressure in gas-phase cleaning systems. Thus, relative compositions of gas mixtures can be varied over a wide range, with the resulting chemistries being able to be adjusted for particular applications. Furthermore, the volatility of certain reaction products can be maximized. Also, at reduced pressures it is possible for the reactants to penetrate narrow openings and to effectively remove unwanted materials from these openings.

New Process Chemistries: Just about any liquid can be converted to a gas. In addition, many additional gas species are available for improved chemical cleaning applications. It is anticipated that such new chemistries will be employed for the more efficient removal of all types of contaminants.

Sequential, In Situ Processing: Perhaps one of the greatest advantages of gas-phase wafer cleaning is its suitability for integrated or cluster processing. Such capability permits various preprocess cleaning steps for critical applications. This subject is discussed more completely in Section 6.7.

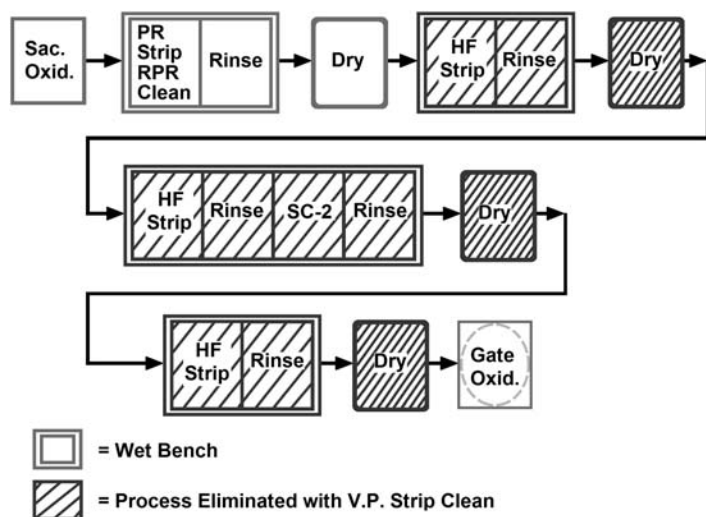


FIGURE 6.1-2 An example of how gas-phase cleaning can replace several liquid cleaning, rinsing, and drying steps in a typical wafer processing sequence for gate oxide formation [39]. *Sac. Oxid.* is sacrificial oxidation and *Gate Oxid.* is gate oxidation. (Courtesy of Advantage Production Technology.)

6.2 CHEMISTRY AND MECHANISMS

An objective of any wafer cleaning operation is to selectively remove from a wafer surface elements and compounds other than those that are purposely deposited there. These undesirable impurities are commonly referred to as either physical (particles) or chemical contaminants. This section considers the mechanisms of dry wafer cleaning, focusing on the removal of chemical contaminants, which can appear on the Si surface in atomic, ionic, or molecular form. Their removal during wet wafer cleaning is accomplished by selective reaction causing either dissolution of the chemical contaminant in the solvent or its conversion into a soluble compound.

The analog of making a product soluble in a liquid phase is to make it volatile in a gas phase. The fundamental steps for contaminant removal are transport of a gas-phase reactant to a surface, adsorption of the species on the surface, reaction with the contaminant to form a product, desorption of the product, and transport away from the surface. Diffusivities in a gas are of the order of $0.1 \text{ cm}^2/\text{s}$ or 10^4 times greater than those in a liquid. An added benefit is that gases have no surface tension. There is a lower resistance for transporting gaseous reactants and products to and from surfaces and gases can “wet” any surface. Molecular flux scales with density, however, so the reactant concentration striking a surface per unit area per unit time in a liquid is of the order of 10^3 times greater than that in a gas. In addition, there is no analog of a

TABLE 6.2-2 Sources of Energy to Activate Gas–Solid Surface Reactions

Energy Source	Implementation	Concerns
Heat	Furnace, rapid thermal processing	Thermal budget side reactions (e.g., substrate etching)
Light	IR, UV	Surface damage
Plasma	Direct, indirect (remote or downstream)	Surface damage, contamination
Particle Beam	Electron, ion, ion, and hot neutrals reactive	Surface damage, complex equipment

liquid solvation shell in a gas, which provides another driving force for product removal. Increasing reaction rates to compensate for the low flux can be accomplished by heating or adding activating species to the gas phase. Sources of activation energy for gas/solid surface reactions, which include heat, light, plasmas, and particle beams, are summarized in [Table 6.2-2](#).

Thermal energy is effective because chemical reactions require a minimum energy to form products. Often there is little thermal budget allowed for cleaning steps in device fabrication and side reactions are a concern, therefore making room temperature processing desirable. Using light reduces the thermal energy needed. Light breaks chemical bonds either directly on a surface or in the gas phase to make reactive fragments, which contain unpaired electrons called radicals that are highly reactive. Plasmas and particle beams contain radicals and electrons that break chemical bonds and generate ions with sufficient kinetic energies to sputter surfaces and activate reactions.

[Fig. 6.2-3](#) illustrates schematically the mechanisms involved in physical and chemical gas-phase cleaning. Physical removal of contaminants occurs via momentum transfer between ions accelerated toward a surface and the contaminant species. The kinetic energy of an accelerated ion is sufficient to break the bond holding the contaminant on a surface. The Si substrate can be eroded as well in this process, and the relative removal rate of the contaminant and Si surface atoms depends on the mass, energy, and direction of the incoming ions. This lack of selectivity can create additional problems related to sputter etching of materials that are purposely formed on the surface and meant to remain there. Redeposition of the sputtered species can also create problems, particularly in the case of large-diameter wafers, narrow geometries, and small mean free paths for gas molecules.

In contrast, chemical removal offers better selectivity than physical sputtering, because a gas-phase species is chosen because of its reactivity with

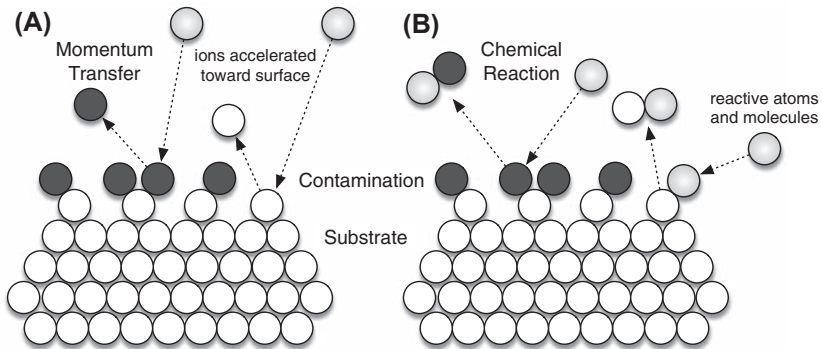


FIGURE 6.2-3 Schematic diagrams of the mechanisms involved in physical and chemical gas-phase cleaning; (A) momentum transfer and (B) chemical reaction mechanisms are depicted. (Used with permission from Anthony Muscat.)

a specific contaminant to make a volatile product, as depicted in Fig. 6.2-3B. This reduces the risk of contamination by redeposition and of incurring surface damage. The chemically reactive species are either mixed in the process gas or are generated by thermal decomposition, photolysis, electron impact, or attachment reactions in plasma. The reaction products with the contaminant are either immediately removed from the surface because the temperature is above the desorption temperature or conditions are created to facilitate this volatilization process by increasing the temperature and reducing the pressure. The latter is a common requirement for gas-phase removal of some metallic contaminants. In a Langmuir–Hinshelwood mechanism, the reactive species first physically or chemically adsorb on substrate sites and through a combination of surface diffusion and reaction produce volatile products. Alternatively, in the Rideal–Eley mechanism, a gas-phase species reacts directly with either a substrate atom or contaminant producing volatile products. Both models are outlined in the study by Masel [40].

The volatilization of metallic species involves chemical and physical phenomena specific to this process, the same phenomena are not necessarily encountered in the case of gas-phase removal of organic species and native/chemical oxides. These effects are illustrated in Fig. 6.2-3B. First, the Si surface is exposed to chemically reactive species that form potentially volatile compounds with metallic contaminants of interest in Si processing. Chlorine is typically the element of choice for this application because metal chlorides are generally more volatile than other metal halides. The metal reaction process must be carried out at low temperature ($<200^{\circ}\text{C}$) to minimize penetration of the metals into Si and to avoid etching of the Si surface by Cl atoms. A Cl_2 partial pressure of 1 Torr illuminated with UV light produces Cl atoms that etch a Si surface above 150°C forming silicon tetrachloride (SiCl_4), which is

volatile, but causes surface roughness [41]. Because SiCl_4 is more volatile than many chlorides of metals that contaminate a Si surface, discussed in Section 6.5 and illustrated in Fig. 6.5-32, it is difficult to prevent volatilization of Si in an environment in which these metal contaminants also are volatile. Depending on the application, slight Si etching might not be a problem as long as it is uniform and does not roughen the surface. A beneficial aspect of Si etching is that along with the stream of volatile SiCl_4 molecules leaving the substrate, some metal chlorides are likely to be lifted off the surface [42], as indicated in Fig. 6.2-3B. A disadvantage of using Cl is the formation of HCl when the process gas comes in contact with moisture, which complicates handling of the effluent. An alternative is to use reactants that remove metal atoms via the formation of highly volatile organometallic compounds. The process is essentially the reverse of a deposition reaction, in which bare ligands coordinate with a metal atom forming a large molecule that has a high vapor pressure.

In Section 6.3.5, the mechanisms are discussed for the chemical removal of contaminants from surfaces using HF/ H_2O vapor (for removing SiO_2), O_3 and UV/ O_3 (for removing organics), and photo-initiated processes (for removing metals and surface preparation). These processes are chosen because they are used, although in a limited extent, for high-volume manufacturing of logic and memory devices. Also these processes, with some modification, are relevant for surfaces that contain Ge.

A number of other gases have been evaluated during the past 25 years for cleaning or etching Si and SiO_2 surfaces. These include halogen fluorides, such as ClF_3 , BrF_3 , and IF_3 [43], and other fluorides including NF_3 , BF_3 , PF_3 , and PF_5 [44]. Some of the procedures involve excitation, such as plasma, whereas others do not. As of this date, few of these alternative processes have been used for high-volume device manufacturing applications.

6.3 REMOVAL OF SILICON OXIDES WITH HF VAPOR

Among gas-phase cleaning chemistries outlined in Table 6.2-2, HF vapor treatments carried out for the purpose of SiO_2 removal are certainly the best explored. The original paper describing vapor HF removal of SiO_2 was written by Holmes and Snell in 1966 [45]. In their experiments, the authors suspended an oxidized Si wafer above an aqueous solution of HF. The solution is agitated by dry Ar, which also helped to transport the HF/ H_2O vapors to the wafer. The authors observed that the rate of attack of the SiO_2 by the vapor is comparable with that in the liquid bath. Oxide etching rate as a function of HF concentration and temperature is shown in Fig. 6.3-4. Two other key results from this early work are that the reaction rate followed the familiar volcano plot, peaking at 20–30°C and rapidly falling to zero at higher ($\sim 40^\circ\text{C}$) temperatures. They also suggested that surface condensation is important. As

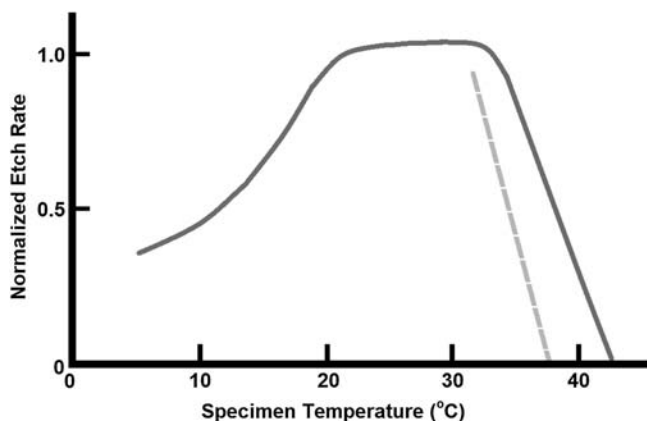
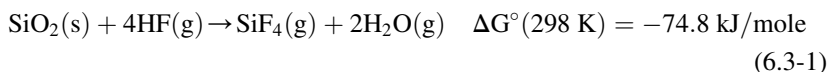


FIGURE 6.3-4 SiO₂ etching rates by HF vapor at 24°C as a function of temperature [45]. Solid line: 40 wt% acid and broken line: dilute acid. (Reprinted from P.J. Holmes, L.E. Snell, *A vapour etching technique for the photolithography of silicon dioxide*, *Microelectron. Reliab.* 5 337. Copyright (1996), with permission from Elsevier and *Microelectronics and Reliability*.)

will be noted later in this chapter, the mechanism for HF etching of SiO₂ involves a requirement for H₂O availability before etching begins, even though H₂O is a product of the reaction. An interesting application related to this effect was reported in 1977 by Bersin and Reichelderfer [46]. In this case, the SiO₂ etching is carried out selectively under a negative photoresist layer, which contained enough H₂O to initiate the reaction. In areas with no resist, SiO₂ etching did not occur.

The overall reaction between SiO₂ and HF gas producing SiF₄ and H₂O is thermodynamically favorable:



where ΔG is the change in Gibb's free energy. An additional polar molecule capable of making a hydrogen bond (H-bond) with HF is needed to lower the kinetic barrier of the reaction, because HF alone does not etch SiO₂. The polar molecule can either be intentionally added to the gas phase or be present on the surface. Water vapor is often used and is autocatalytic because it is a reaction product. Fig. 6.3-5 shows the typical time response of the ellipsometric parameter ψ , the change in film thickness during etching of a thermal SiO₂ film with a low-pressure mixture of HF and H₂O vapor. The parameter ψ , which undergoes a sinusoidal variation as the SiO₂ thickness changes, displays three distinct behaviors as a function of reaction time, starting when HF enters the reactor. These stages are termed initiation, bulk etching, and termination. The characteristics of each stage are discussed in turn in the following sections.

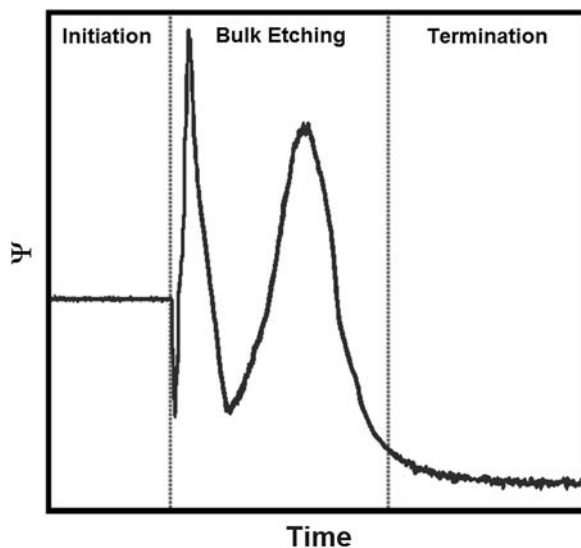


FIGURE 6.3-5 Typical response of the ellipsometric parameter ψ during etching of a thermal SiO_2 film with a low-pressure mixture of HF and H_2O vapor, showing three stages with respect to time. (Used with permission from Anthony Muscat.)

6.3.1 Initiation

In addition to H_2O vapor, molecules such as alcohols and carboxylic acids in the gas phase are effective at lowering the kinetic barrier of the etching reaction. Moreover, moieties on the surface of the oxide such as silanol (SiOH), hydroxylated boron (BOH), and phosphorus (POH) also enhance the reaction [47]. The interaction of HF with these groups and with surface siloxane (Si-O-Si) moieties produces a delay in etching called the induction or initiation time, which is of the order of 10 s. As shown in Fig. 6.3-5, the SiO_2 thickness is constant during this time, indicating that no etching occurs. The surface chemistry is far from dormant, however, because temperature, pressure, HF concentration, vapor mixture and concentration [48], additives such as HCl [49], and oxide type [50,51] affect the induction time. Although no definitive spectroscopic evidence has been presented in published literature to show the sequence of reaction events, many experimental results have been explained using a liquid-phase model.

That H_2O is required for initiation suggests that etching occurs in a thin liquidlike layer on the surface of the SiO_2 [52]. The induction period is the time required for H_2O and HF to condense on the surface and form active etching species. The agreement between experimental data and the thermodynamic model predicting a condensed layer will form plus the appearance of residues on SiO_2 surfaces after etching support this interpretation. Clearly, some etching

conditions support the formation of a liquid layer on the surface. This is not a necessary condition, however, as a series of studies done in 1995 showed that etching can occur without a liquid layer on the surface [51,53]. Control of the liquid layer affects SiO_2 etching selectivity [54] and can be important for removing oxides from other Group IV materials, such as Ge, and Group III-V materials, which are known to be sensitive to H_2O .

Etching does not occur without H_2O adsorption on the surface [50], however, the layer is present only during etching. In situ Fourier transform infrared (FTIR) spectroscopy showed that adsorbed H_2O appears on the surface of thermal SiO_2 films when HF is introduced to start the etching and decreased when the film is removed [55]. Water is not adsorbed on the surface before etching even though the partial pressure of H_2O in the gas phase is the same as when HF is introduced. These results are corroborated by mass spectrometer measurements during etching [50]. Similar conclusions can be made for HF, namely that HF does not etch thermal SiO_2 films without H_2O vapor present. First principles molecular modeling, starting with a hydroxylated surface (SiOH), concluded that Coulombic attraction between polarized atoms, in this case H in an HF molecule and O in the oxide matrix is responsible for initiating the etching [56]. Another first principles study showed that the concerted attack of HF and H_2O molecules on the Si—O bond of surface silanol (SiOH) groups is a low-energy etching pathway [57]. A gas-phase H_2O molecule serves to catalyze the reaction by donating one H to a surface SiO—H and accepting one H from HF. Experimental evidence for a coordinated effect based on in situ FTIR spectroscopy showed that HF and H_2O molecules formed a cluster in the gas phase under low-pressure (<200 Torr) conditions [55]. The initiation time correlated with the cluster concentration, suggesting that the HF— H_2O complex adsorbs on a SiO_2 surface more strongly than either molecule alone. Consequently, water acts as a carrier of HF molecules, providing a means for HF to H-bond to SiO_2 and etch the surface.

6.3.2 Bulk Etching

Etching of the SiO_2 initiates once the surface is covered with adsorbed HF and H_2O molecules. This is detected as a change in film thickness by an ellipsometer, represented by the parameter ψ in Fig. 6.3-5. Oxide films that start etching immediately with no induction period contain enough adsorbed H_2O that HF sticks to the surface. Silicate glass films that contain boron (B) or phosphorus (P) typically show this behavior [47]. The response surface for etching thermal SiO_2 films as a function of the H_2O and HF flow rates in a reactor at 34°C and 100 Torr is shown in Fig. 6.3-6. The total flow rate is fixed at 563 sccm (standard cubic centimeters) with makeup N_2 and is used to calculate the partial pressures of HF and H_2O . The broad span of the reaction

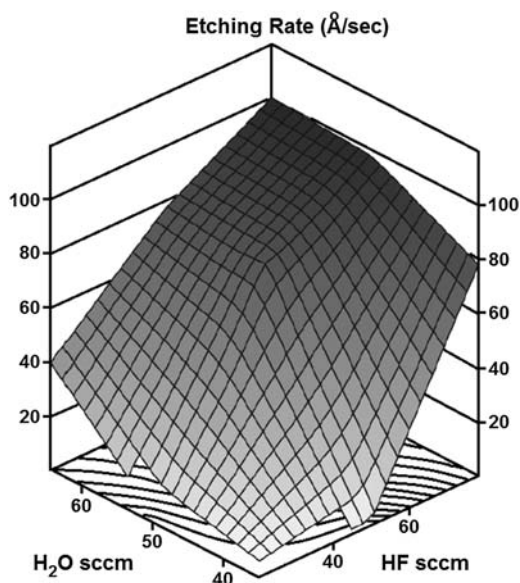


FIGURE 6.3-6 Response surface plot of thermal SiO_2 etching rate in gas-phase HF and H_2O vapor mixtures at 34°C and 100 Torr [55]. (Used with permission from Gerardo Montano-Miranda.)

rate from $\sim 1 \text{ Å/s}$ at 20 sccm (4 Torr) of HF and 35 sccm (6 Torr) of H_2O to 100 Å/s at 85 sccm (15 Torr) of HF and 65 sccm (12 Torr) of H_2O is typical of etching thermal SiO_2 films. In the experiments, all of the etching processes start as gas/solid reactions even though the surface is exposed to H_2O vapor for ~ 10 min before introducing HF gas, because H_2O is not detected on the surface by FTIR spectroscopy. This indicates that for subatmospheric pressures at 34°C , a liquid H_2O layer is not a prerequisite for etching, but rather a consequence of the reaction. Water did not buildup on the surface at low etching rates, whereas several layers of H_2O , which produced a liquid film on the surface at high etching rates.

The stability of this liquid film depended on the rate of evaporation, which is a function of the gas-phase concentration of the reactants, pressure, and temperature. Besides kinetics, the H_2O layer influences the etching rate through the dilution thermodynamics of HF in the liquidlike layer and the transport of species across the gas and liquid boundary and reaction layers above the surface. The influence of transport can be seen in the initiation plateau that starts to develop at high etching rates. This plateau is caused partly because of the additional resistance offered by the increased production of H_2O at higher etching rates, developing into a proportionally thicker layer that opposes the diffusion of reactive species to the surface of the sample. This influence is not observed at low etching rates because of the absence of a H_2O layer. Without a detectable H_2O layer observed with FTIR, the etching rates

are almost two orders of magnitude lower than those achieved with a detectable H_2O layer. This is likely the result of a change in the active etching species attacking SiO_2 in both the gas/solid regime and the gas/liquid/solid regime. In the gas/solid regime, the reaction takes place through gas-phase species, including gas-phase HF, H_2O , and the HF/ H_2O cluster. In the gas/solid/liquid regime, the etching species are similar to those found in liquid-phase etching, albeit different because of the higher concentrations of HF that can be achieved in a gas/liquid/solid process.

The most successful quantitative model of gas-phase HF/ H_2O etching of SiO_2 in the bulk etching regime where a H_2O layer is present on the surface of the oxide film is the work by Helms and Deal [52]. Etching rates of thermal oxide films in a liquid-phase HF/ H_2O solution up to approximately $150 \text{ \AA/s}$, as shown in Fig. 6.3-7 [58–60], are similar to those in an HF/ H_2O gas-phase regime, as shown in Fig. 6.3-8. This suggests that the mechanism of bulk etching is similar to liquid-phase etching. Helms and Deal used this idea to develop a steady-state model that predicts the stability of a condensed liquid in equilibrium with a vapor. Although it is now known that a condensed liquid layer is not a necessary condition for etching, the Helms–Deal model, nevertheless, correctly predicts rates in the bulk etching regime when a liquid layer is present on the surface.

The steps for SiO_2 removal start with transport of HF and H_2O from a vapor phase into the boundary layer, as shown in Fig. 6.3-9. The hydrodynamic boundary layer is defined by the interfacial region between the vapor and liquid phases where the gas velocity changes from the free stream value to zero at the liquid

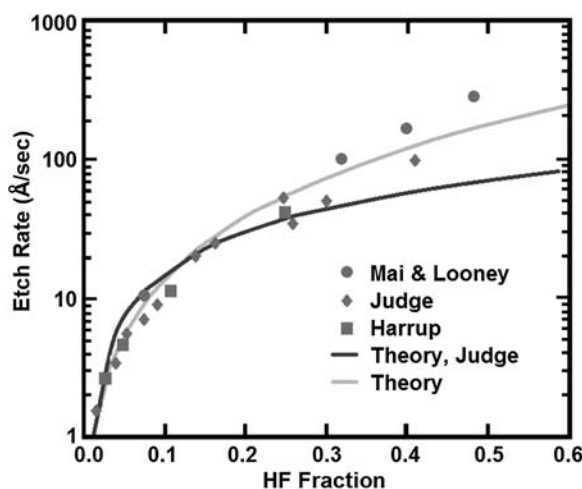


FIGURE 6.3-7 Thermal SiO_2 etching rate data as a function of HF concentration (wt%) in HF– H_2O mixtures based on early results from Mai and Looney [58], Judge [59], and Harrap [60]. The dashed line is from Judge and the solid line an improved fit of the data. (Courtesy of Bruce E. Deal and C. Robert Helms.)

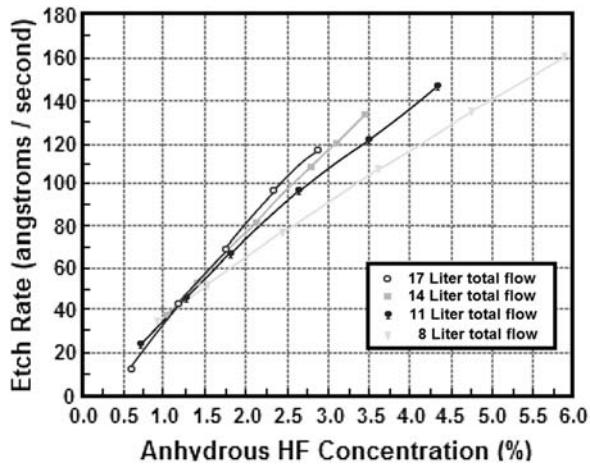


FIGURE 6.3-8 Thermal SiO_2 etching rates at various anhydrous HF concentrations at 25°C for various flow rates in a commercially available vapor cleaning systems. (Courtesy of FSI International.)

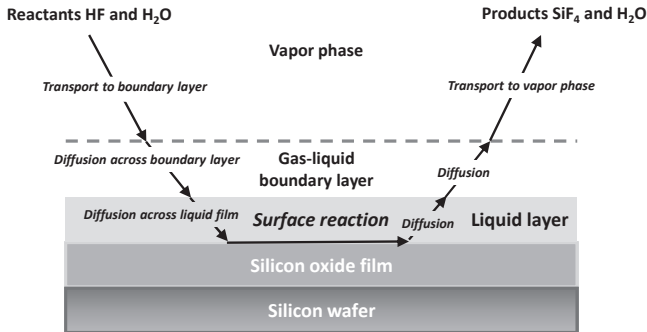


FIGURE 6.3-9 Overview of transport and reaction processes in SiO_2 etching using gas-phase mixtures of HF and H_2O vapor. (Used with permission from Anthony Muscat.)

surface. The reactants diffuse across the boundary, dissolve in the liquid layer, subsequently diffuse across the liquid layer, and finally react at the surface. The etching reaction removes the SiO_x film, producing hydrofluorosilicic acid (H_2SiF_6) and H_2O (product H_2O). The products undergo transport because of the concentration driving force back to the vapor phase by diffusion and vaporization at the gas–liquid boundary layer. The products are carried away from the surface by the flow of the vapor stream. Undesired secondary reactions can cause residue formation not encountered with aqueous processing. For example, there is a driving force for disproportionation of H_2SiF_6 into silicon tetrafluoride (SiF_4) and HF at the gas–liquid interface.



At high etching rates, residues containing nonvolatile H_2SiF_6 can form and will remain on the surface after pumping out the gas, unless immediately rinsed with liquid H_2O . Mass transport limitations have been observed in the bulk etching regime and modeled using classic stagnation flow for high etching rates [61].

In the Helms–Deal model equilibrium, partial pressures of HF and H_2O are used to determine the stability conditions for a liquid layer, which is used to determine etching rates by comparison with liquid-phase etching models [52]. This approach assumes that the system is at steady state with a flux of reactants, F , at the surface given by the ideal gas collision frequency

$$\Phi_i = \frac{P_i}{\sqrt{2\pi m_i kT}} \quad (6.3-3)$$

where P_i is the partial pressure of reactant i , m_i is the mass of reactant i , and kT is the energy associated with temperature T . The flux of reactants leaving the surface is given by the appropriate reactant vapor pressures for the liquid phase. This is only valid if the sticking probabilities for the HF and H_2O are nearly equal. This is indeed the case, as described below. Comparing masses, $m_{\text{HF}} = 1.05m_{\text{H}_2\text{O}}$, so assuming that the fluxes of each are proportional to their partial pressures leads to minimal error. For these conditions, the net steady-state flux into the condensed layer is

$$\Phi_{\text{H}_2\text{O}} = P_{\text{H}_2\text{O}} - P_{\text{H}_2\text{O}}^0(T, [\text{HF}]) \quad (6.3-4)$$

$$\Phi_{\text{HF}} = P_{\text{HF}} - P_{\text{HF}}^0(T, [\text{HF}]) \quad (6.3-5)$$

where the Φ are the fluxes, P is the pressure, T is the surface temperature, and $[\text{HF}]$ is the HF concentration in the condensed layer. $P_{\text{H}_2\text{O}}$ and P_{HF} are the gas-phase partial pressures right above the surface and $P_{\text{H}_2\text{O}}^0$ and P_{HF}^0 are the equilibrium vapor pressures at sample temperature T and condensed layer composition $[\text{HF}]$. The HF concentration at steady state is determined by the ratio of the net HF flux to the total reactant flux

$$[\text{HF}] = \frac{P_{\text{HF}} - P_{\text{HF}}^0(T, [\text{HF}])}{P_{\text{HF}} - P_{\text{HF}}^0(T, [\text{HF}]) + P_{\text{H}_2\text{O}} - P_{\text{H}_2\text{O}}^0(T, [\text{HF}])} \quad (6.3-6)$$

For a given set of experimental conditions, this expression can be solved, provided values for the equilibrium vapor pressures are available. Fortunately, there are extensive data in the literature on vapor/liquid equilibria for HF/ H_2O mixtures. The most complete reference on HF/ H_2O vapor/liquid equilibria is that of Munter et al. [62,63]; their data are shown in Fig. 6.3-10 where the equilibrium vapor pressures, including the values for pure HF, are plotted versus liquid composition for a series of temperatures. Additional data on the high HF concentration limit, at 50°C , are also shown [64]. For low-HF

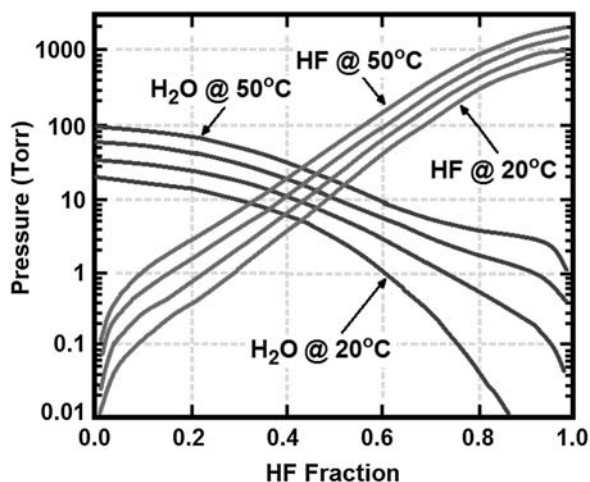


FIGURE 6.3-10 Vapor pressures of HF and H₂O with changes to the mole fraction of HF at various temperatures [52]. (Reprinted with permission from C.R. Helms, B.E. Deal, *J. Vac. Sci. Technol. A10* (4) (1992) 806. Copyright 1992, American Institute of Physics.)

concentrations, the vapor pressure of HF is very low so that the HF fraction in the liquid phase will be much larger than that in the gas-phase. The converse is true for the limit of high HF concentrations. This in part explains etching with anhydrous HF, H₂O present or created by the reaction can be retained in the liquid because of the low H₂O vapor pressure at high HF concentrations. If the sticking probabilities are equal for all concentrations, the point where the gas-phase and liquid-phase partial pressures are equal will be the azeotrope. This occurs at approximately 39 wt% HF [62,63], which is near the equal partial pressure points of both H₂O and HF. This supports the assumption of equal liquid sticking probabilities, as does the experimental observation that HF and H₂O molecules cluster in the gas-phase. The data in Fig. 6.3-10 are modeled using a near-regular solution approach based on van Laar equations for liquid-phase activity coefficients. The model has been refit from the original work by Helms and Deal [52] to recent data for the high HF partial pressure limit and all subsequent curves reflect this fit [64].

At a given temperature, condensation will form on the surface, producing a liquid layer that is in equilibrium with its vapor, but only if the combination of gas-phase partial pressures is high enough so that the flux of molecules impinging on the surface is greater than that leaving by evaporation. This is represented by Eq. (6.3-6) where both the numerator and denominator must be greater than zero. The result is shown in Fig. 6.3-11 for 20°C where condensation will occur on the high-pressure side of the line (upper right) and where no condensation is expected on the low-pressure side of the line (lower left).

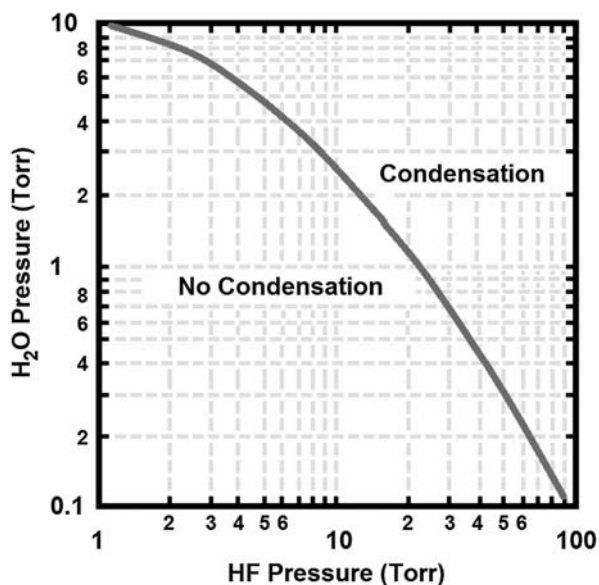


FIGURE 6.3-11 Partial pressures of HF and H₂O indicating ranges corresponding to stable condensation at 20°C [52]. (Reprinted with permission from C.R. Helms, B.E. Deal, *J. Vac. Sci. Technol. A*10 (4) (1992) 806. Copyright 1992, American Institute of Physics.)

The Helms–Deal analysis assumed that no etching can occur in the noncondensed regime. However, studies show that etching is possible at the gas/solid regime, albeit at a lower rate than in the gas/liquid/solid regime [53,65]. The Helms–Deal model is still valid but applies only when the partial pressure is high enough to support a stable liquid layer. The problem of nucleating a condensed film on a surface is neglected in the Helms–Deal analysis. The condensation explanation is undoubtedly justified because of the hydrophilic nature of the H₂O monolayer produced by this process.

6.3.2.1 Etching Rate

Etching rates can be predicted for the gas-phase process and compared with liquid-phase etching rates at the same HF concentration. Adsorption and desorption kinetics and mass transport are neglected as are the depletion of HF during etching and the generation of byproducts because these steps are likely very different for a gas-phase process compared with a liquid-phase process. The HF concentration in a liquid layer as a function of the partial pressures of HF and H₂O in the vapor phase is computed using the data in Fig. 6.3-10 and Eq. (6.3-6). The etching rate predicted at this HF concentration is computed using the curve fit to the liquid-phase data shown by the solid line in Fig. 6.3-7. The result is the predicted gas-phase etching rate at 25°C based on a liquid

layer in equilibrium with HF and H_2O partial pressures and is shown in Fig. 6.3-12. This calculation also assumes a negligible pressure drop across any boundary layer or pressure gradient because of gas-phase diffusion limitations.

Experimental etching rates in the gas-phase range from a few $\text{\AA}/\text{s}$ for dilute HF concentration to hundreds of $\text{\AA}/\text{s}$ for high HF concentrations, as shown in Fig. 6.3-7 for 34°C . The predicted rates in Fig. 6.3-12 for 25°C show that the values are in the range expected. A quantitative comparison at 25°C using the results of Deal et al. [39,66] where vapor pressure data are available is shown in Fig. 6.3-13. The fit of the liquid-phase etching data in Fig. 6.3-7 (solid line) is shown for comparison. The agreement is reasonably good, but the model appears to underpredict the HF concentration in the liquid layer necessary to achieve a specific etching rate at low concentrations and overpredict the HF concentration necessary to equate gas and liquid etching rates at high concentrations. The latter effect may be because of incomplete mixing in the liquid layer generated by the gas-phase process because the liquid condensation rates are measured for stirred conditions and the buildup of reaction products. The deviation at low concentration is likely because of the inaccuracy of the vapor pressure model for low HF concentrations where no data are available.

An example of the amount of thermal SiO_2 etched as a function of time is presented in Fig. 6.3-14. The initiation times to start etching is given by the

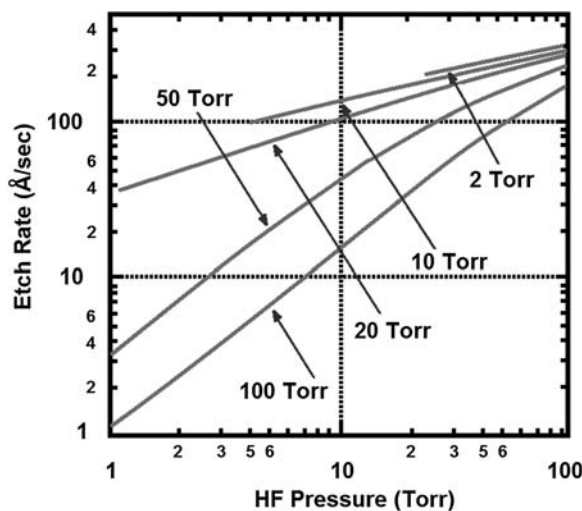


FIGURE 6.3-12 Predicted SiO_2 etching rates at 25°C as a function of HF partial pressures at the indicated H_2O pressures, obtained by combining the equilibrium model for gas-phase etching with the liquid-phase etching rate data from Fig. 6.3-7 [52]. (Reprinted with permission from C.R. Helms, B.E. Deal, *J. Vac. Sci. Technol. A* 10 (4) (1992) 806. Copyright 1992, American Institute of Physics.)

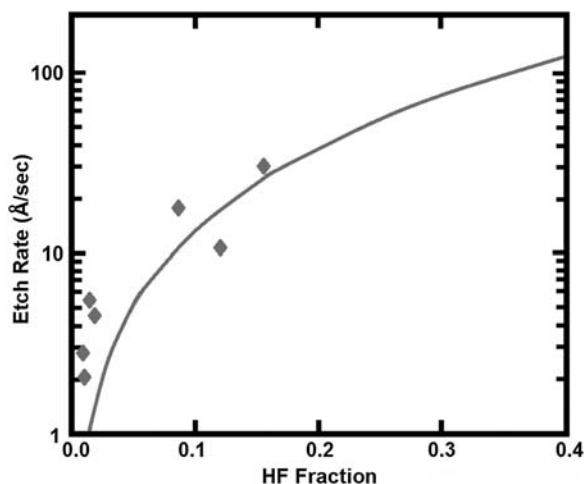


FIGURE 6.3-13 The points show the gas-phase SiO_2 etching rates at 25°C from Refs. [39,66] versus the equivalent HF fraction in a liquid-phase calculated using Eq. (6.3-6). The solid line is the same as in Fig. 6.3-7 and shows the liquid-phase etching rate [52]. (Reprinted with permission from C.R. Helms, B.E. Deal, *J. Vac. Sci. Technol. A10* (4) (1992) 806. Copyright 1992, American Institute of Physics.)

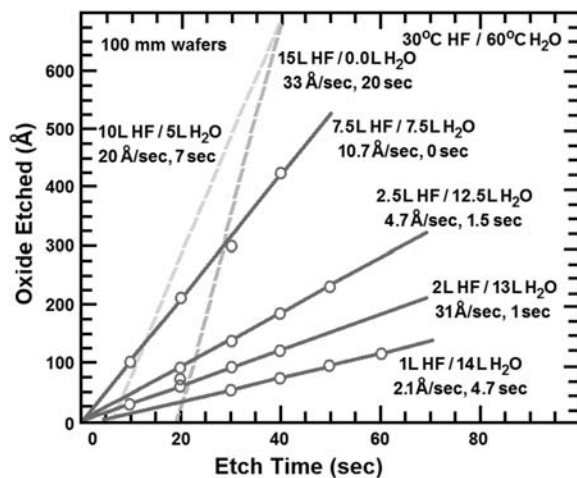


FIGURE 6.3-14 The amount of thermal SiO_2 (oxide) removed versus time for various ratios of azeotropic $\text{HF}/\text{H}_2\text{O}$ solutions (38.4 wt% HF) to H_2O vapor. The wafer temperature is 25°C , the azeotrope temperature is 30°C , and the H_2O vaporizer temperature is 60°C . The flow rate is 15 L/min [39]. The etching rate is calculated from the slopes of the lines in $\text{\AA}/\text{s}$, and the initiation time obtained from the time intercept is shown in seconds for each ratio studied. (Used with permission from *Solid State Technology*.)

TABLE 6.3-3 Typical Results for Vapor Phase Processes [168]

Vapor Composition	TEOS Oxide Etching Rate (Å/s)	Thermal Oxide Etching Rate (Å/s)
Azeotropic HF	7.8	48
Azeotropic HF + H ₂ O	5.4	3.2
Azeotropic HF + HCl	(Not measured)	3.3
Anhydrous HF + IPA	6	1.5

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time intercepts, as a function of HF/H₂O vapor ratio, which varies from 0 to 20 s. An estimation of HF concentration in the condensed aqueous layer for a particular HF/H₂O vapor ratio can be made by comparing etching rates with those of Fig. 6.3-7 for aqueous solutions [39]. For example, a 4.7 Å/s etching rate obtained by mixing 2.5 L/minute of azeotropic HF with 12.5 L/minute of H₂O corresponds to about a 5 wt% HF aqueous solution. Estimation of the HF concentration in a liquid layer can be used to determine etching selectivity differences between different types of silicon oxides and silicate glass films.

For partial etching of blanket SiO₂ layers or selective etching of oxides, it is important to have process repeatability; etching as uniformly as possible across the wafer and wafer-to-wafer. Kao et al. [67] observed improvement in SiO₂ across the wafer etch uniformity for O₃ pretreated wafers, confirming that enough organic contamination is present on the SiO₂ surface to affect etching uniformity [68]. However, etching rate repeatability from wafer-to-wafer is not affected appreciably. Other process variables that can affect gas-phase etch characteristics include temperature, chamber pressure, and any variable affecting composition of the condensed HF aqueous layer, such as gas flow rates, vaporizer composition and temperature, and type of solvent. Table 6.3-3 shows typical process results.

6.3.2.2 Silicon Oxide—Chemical Oxide, Native Oxide, Thermal Oxide, and Deposited

A majority of applications involving gas-phase HF etching of SiO₂ relate to complete removal of a thick or thin SiO₂ layer. After this removal, and after various aqueous cleaning treatments of “stripped” Si surfaces, a thin (5–15 Å) SiO_x layer typically remains. Thus, the important subject of the formation of these so-called native or chemical silicon oxide, including native oxide regrowth, arises [69–74]. These layers vary in composition, depending on the treatment, often contain SiO₂, organic compounds, various anions such as sulfates, and other impurities. Many device structures are affected critically by the presence or absence of this thin layer of questionable composition. These

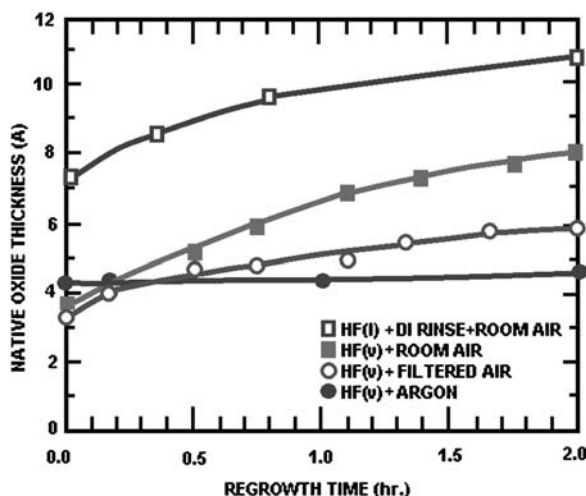


FIGURE 6.3-15 An example of native SiO_2 regrowth at 25°C in various ambient atmospheres following vapor and liquid HF stripping of thermal oxide [77]. (Reprinted from *Microcontamination*. Copyright 1992, Tom Cheyney, Los Angeles, CA. Used with permission.)

structures include interconnect contacts, polySi bipolar emitters, epitaxial Si layers, and MOS gate SiO_2 . Practical aspects of precleaning effects on device properties are discussed in Section 6.6.

It is very difficult to characterize chemical oxides. Various techniques have been used to determine their thickness and composition, such as XPS (X-ray photoelectron spectroscopy), Auger electron spectroscopy (AES), secondary ion mass spectroscopy (SIMS), and spectroscopic ellipsometry. Surface and embedded impurities have been characterized by total reflection X-ray fluorescence (TXRF) and surface morphology has been characterized by electron energy loss spectroscopy, atomic force microscopy, and scanning tunneling microscopy. Some of these techniques are discussed in Chapter 13.

Kinetics of film growth have been investigated and typical results are presented in Fig. 6.3-15, where it is noted that after a conventional aqueous HF etch and DIW rinse, the chemical SiO_2 thickness is $7\text{--}8\text{ \AA}$. For a gas-phase HF treatment, however, the thickness is about $4\text{ \AA}$ for this particular substrate, as measured by an ellipsometer. If both wafers are exposed to an ambient atmosphere, a steady thickness increase is observed over time, but is much less for exposure to either charcoal-filtered air or Ar (argon). This has led to the conclusion that much of the so-called reoxidation is really because of condensation of organic and other impurities on the surface [75,76].

One of the device fabrication requirements for removing native/chemical SiO_2 from contact openings is to minimize the attack on the surrounding dielectric and ideally, not to etch it at all. It was found by Miki et al. that anhydrous HF can remove native oxide layers without etching adjacent thermal SiO_2 [54]. In their work, a gas-phase HF critical concentration is

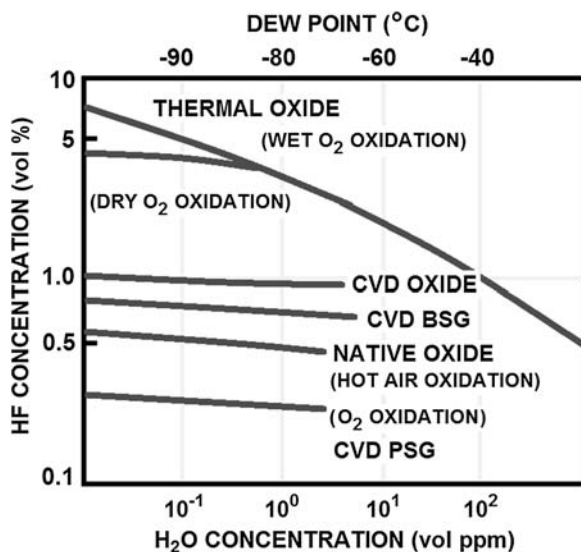


FIGURE 6.3-16 Relationship of HF critical concentration for oxide etching and the moisture level (in ppm — parts per million) at 25°C for various dielectric films [54]. (Reprinted with permission from Miki et al., *IEEE Transactions on Electron Devices* 37 (1) (1990) 107. Copyright 1990, IEEE.)

found, below which no etching is observed. This critical concentration depends on residual H_2O , temperature, and flow rate and the nature of the SiO_2 being etched. They suggested that H_2O is necessary for etching to proceed and that hydrated surfaces, such as chemical or native SiO_2 , will therefore have lower HF partial pressure thresholds, whereas drier surfaces, such as thermal SiO_2 , will have higher thresholds. Once the reaction is initiated at sufficiently high HF pressure, enough H_2O will be produced to allow the reaction to continue. Data that demonstrate such selective oxide etching are presented in Fig. 6.3-16. It is also observed that a selective etching process can be achieved using the vapor HF/ H_2O reaction. This is accomplished at conditions that have a delay time of 10 s or more before etching of the thermal oxide begins. Because the condensed aqueous HF solution has not yet formed, the gaseous HF will etch the native oxide but not the thermal oxide. This phenomenon is illustrated in Fig. 6.3-17, where an 8 Å native SiO_2 is removed before etching of the thermal SiO_2 commences.

Various types of dielectric films, usually based on SiO_2 , are used in semiconductor device fabrication [25,77,78]. These films can be deposited by a variety of methods and may contain additives, such as P and B in amounts up to 10 wt%, to form silicate glass films. As a result of doping, silicate film etching rates can vary considerably. For a given etching process, aqueous or vapor, each dielectric film must be characterized. In addition, when two or more dielectrics are present on the same wafer, differential etching rates are

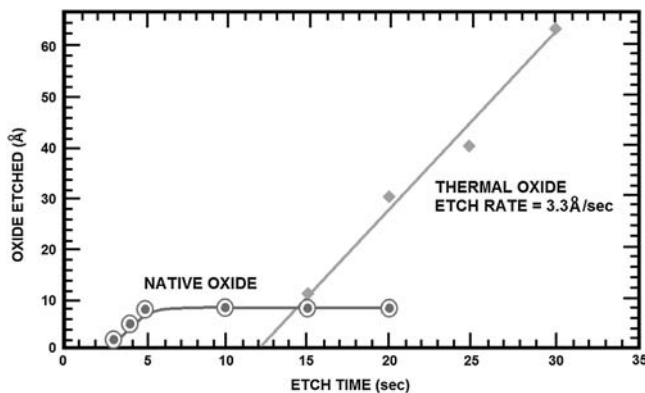


FIGURE 6.3-17 Relationship of etch time and oxide thickness etched. Selective etching of native oxides in the presence of thermal oxides using vapor HF process, indicated by the difference in etching rates. (Courtesy of Bruce E. Deal and C. Robert Helms.)

TABLE 6.3-4 The Relative Etch Selectivity of Various Thin Oxide and Silicate Glass Films Compared With Thermal Oxide at 45°C HF Vaporizer Temperature and 12-s Etch Time

Films	Thickness Etched (Å)	Selectivity
Thermal oxide (SiO ₂)	240	1.0
LTO (SiO _x)	360	1.5
TEOS	550	2.1
PSG (4 wt% P)	776	3.2
BPSG (4 wt% B, 7 wt% P)	1288	5.4

LTO is low-temperature deposited oxide, typically by PECVD (plasma-enhanced CVD) or thermal CVD. TEOS is a precursor used for SiO_x deposition in a PECVD system. PSG is phosphosilicate glass and BPSG is borophosphosilicate glass. Compiled by the authors from data from Advantage and Genus.

often observed. Many times, etching rate differences can be used as advantage for fabricating a particular device structure, whereas in other cases severe problems result. In [Table 6.3-4](#), examples of relative etch amounts for different dielectric films are shown for a given etch time in a gas-phase HF/H₂O system. Similar data have been reported for other vapor etching systems; shown in [Fig. 6.3-18](#).

6.3.3 Termination, Residue Formation, Surface Roughness, and Electrical Performance

The XPS survey spectrum of a Si surface in [Fig. 6.3-19](#), taken after a typical HF/H₂O experiment that stripped the SiO₂, shows that F, O, and a small

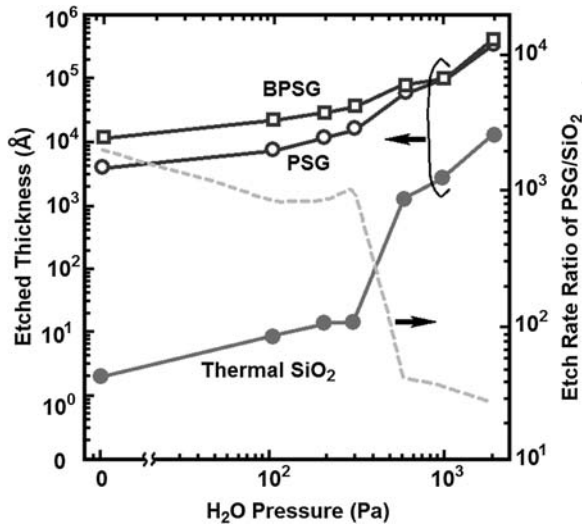


FIGURE 6.3-18 Etched thickness of BPSG, PSG, and thermal SiO₂ versus the partial pressure of H₂O with a fixed HF partial pressure of 600 Pa. Films are exposed to the gas mixture for 30 s [51]. (Reproduced by permission of ECS—The Electrochemical Society from H. Watanabe, H.S. Ohnishi, I. Honma, H. Kitajima, H. Ono, R.J. Wilhelm, A.J.L. Sophie, J. Electrochem. Soc.)

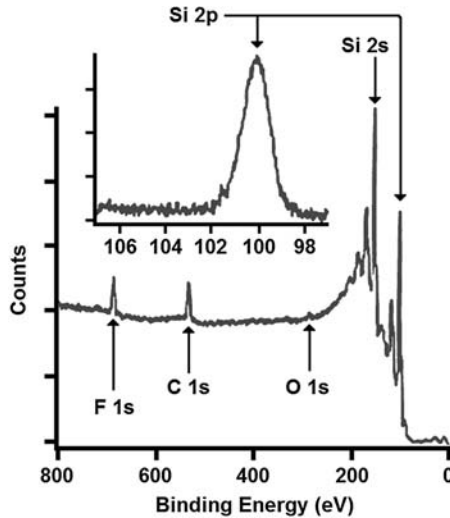


FIGURE 6.3-19 Survey (200 eV) and high-resolution (50 eV) XPS traces of the surface after SiO₂ removal with HF/H₂O vapor [55]. The process completely removed the SiO₂ and shows that F, O, and a small amount of C are present on the Si surface. (Used with permission from Gerardo Montano-Miranda.)

amount of C are present. The inset, from the Si region at 50-eV pass energy, shows only a Si metal peak indicating complete removal of Si suboxides. The survey spectrum shows that the peak at 686 eV is F in the form SiF_x ($x = 1, 2, 3$) and the peak at 532.4 eV is O in the form of silanol (SiOH) groups. The existence of SiOH is corroborated by FTIR data. The maximum amount of residual O as SiOH is typically found at moderate HF and H_2O partial pressures. The minimum amount of O in the layer is typically found at the highest HF concentrations. Gas-phase SiO_2 etching produces higher F surface coverage than liquid-phase processes, which is a consequence of the H-bonded H_2O layer on the surface. Although F-termination is not desirable if the subsequent step is deposition, F atoms can be replaced with other terminating groups that facilitate deposition. Chapter 9 discusses the hydrogen termination of Si surfaces by wet HF treatment.

Refinement of vapor HF etching techniques, so far, not has been able to eliminate problems associated with this process. The most persistent one is solid residues, formed as a product of etching, that remain on the surface. An additional H_2O rinse must be applied following vapor HF exposure to wash these residues off. The positive side using this process as a last treatment is the ability to remove some metallic residues in the form of soluble fluorides [79]. Still, the need for H_2O rinsing seems to defeat the purpose of what is intended to be an entirely gas-phase operation. Also, a dependence of the vapor HF process on the moisture content [80], the degree of hydration of etched SiO_2 , and the amount of surface hydrocarbons can all potentially lead to etch nonuniformity, difficulty controlling the etch selectivity, and very high etching rates resulting in undesired or excessive etching of some oxides of Si, in particular BPSG (borophosphosilicate glass). As a result, moisture content must be precisely controlled to achieve adequate process controllability, as shown in Fig. 6.3-20. Moreover, the need to add moisture to anhydrous gaseous HF to control etching reactions leads to the corrosion of stainless steel gas lines typically used in reduced-pressure processing.

Two disadvantages of gas-phase SiO_2 etching and cleaning are (1) the lack of a liquid-phase solvent and (2) the need to remove reaction products by evaporation rather than dissolution. In some cases, a final H_2O rinse is used [81], but the addition of a wet step at the end of the process defeats the purpose of gas-phase only etching and cleaning. Two gas-phase methods have shown promise for providing residue-free etching and cleaning, one employing HF and H_2O at elevated temperatures [82,83] and one employing HF with methanol or isopropanol as the condensable solvent medium [48,83–85]. If properly controlled, the HF/ CH_3OH gas process suppresses the formation of solid residue on the etched surface, as shown in Fig. 6.3-21, which results in an improved etch selectivity compared with the HF/ H_2O vapor process. At the same time, however, lower SiO_2 etching rates are observed for this process compared with etching with HF/ H_2O vapor.

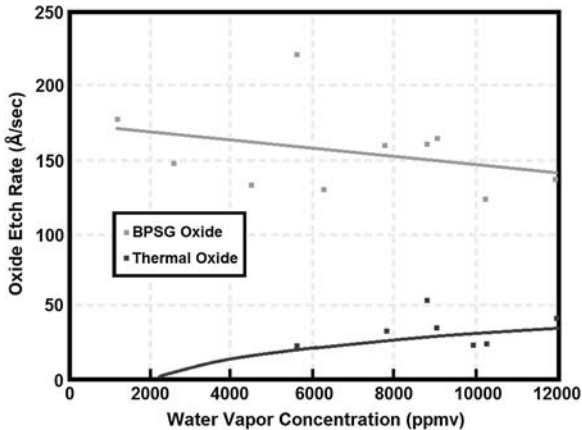


FIGURE 6.3-20 Moisture content must be precisely controlled to achieve adequate process controllability. The added moisture to anhydrous gaseous HF decreases the etching rate of BPSG, but increases the etching rate of thermal SiO₂. (Courtesy of FSI International.)

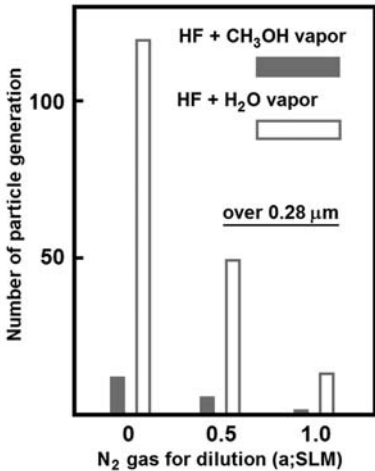


FIGURE 6.3-21 Moisture content must be precisely controlled to achieve low deposition of residue, counted as particles in this experiment [84]. (Reproduced by permission of ECS—The Electrochemical Society from A. Izumi, T. Matsuka, T. Takeuchi, A. Yamano, Second International Symposium on Cleaning Technology in Semiconductor Device Manufacturing.)

The residues that can form during etching and cleaning occur by the competition between reaction product desorption and other side reactions, which can be represented approximately as:



The equilibrium suggests conditions that favor rapid desorption of SiF_4 , shown in Eq. (6.3-2), at high HF pressures. Low H_2O concentrations will produce less H_2SiO_3 . Onishi et al. [82] have shown that increasing the reaction temperature by tens of degrees gives much lower residue concentrations. This likely is related to the greater volatility of SiF_4 at elevated temperatures. Alcohols in place of H_2O as the solvent also have shown improvements in residue formation.

There are other implications of the potentially high reaction product limit associated with gas-phase HF etching and cleaning. First, regarding the etching process, Thomsen [86] has shown that H_2SiF_6 is capable of etching SiO_2 via an intermediate reaction:



In addition, the presence of high concentrations (~ 10 mol%) of H_2SiF_6 in HF/ H_2O solutions significantly reduces the vapor pressures of both HF and H_2O in the solutions. This is a useful mechanism, which stabilizes the condensed liquid-phase at high HF concentrations, but may lead to residue formation.

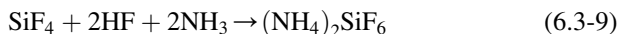
6.3.4 Other Dry Methods of Oxide Etching

Ruzylo et al. investigated UV/ $\text{NF}_3\text{:H}_2\text{:Ar}$ etching of Si oxides [87]. This method is found to be suitable for native/chemical oxide etching, including premetal oxide etch. As the current–voltage (IV) characteristics of Ruzylo's work indicate, this process is able to replace the conventional HF dip applied immediately before contact metallization without compromising the characteristics of Al–Si (aluminum–silicon) ohmic contacts. Similar chemistry previously was activated by ArF excimer laser (193 nm); however, enhanced etching of native oxide was observed [88].

Another development concerns the use of UV exposure in conjunction with anhydrous HF/ CH_3OH chemistry [89]. In this process, the UV irradiation is used to control oxide etching reactions, leading to increased process versatility and improved performance. The UV irradiation is also used to control moisture level on the Si surface. In addition, UV/ O_3 treatment can be applied in situ before native oxide etching to remove hydrocarbons and, hence, to assure adequate etch uniformity.

With aqueous and gas-phase HF etching, thermal oxides are etched at a slower rate than deposited oxides, which are less dense and have a higher moisture content. The etching selectivity can be reversed in which thermal oxide etch at a higher rate than deposited oxides by forming residues on the surface that retard reactant diffusion. The etching process becomes diffusion limited as a consequence of the surface residues, instead of reaction limited as a consequence of the condensed layers. The salt $(\text{NH}_4)_2\text{SiF}_6$ is identified as the surface residue by XPS-determined ratios of N, Si, and F atoms. This residue

deposits after exposing both tetraethylorthosilicate (TEOS) and thermal oxide to gas-phase mixtures of HF and NH₃, a process called chemical oxide removal (COR) [90]. The authors propose that the salt forms when SiO₂ is etched producing SiF₄, which reacts with HF and NH₃ on the surface. The overall reaction is as shown below:



In a plot of the amount etched as a function of process time, deposited oxides initially etch at a higher rate than thermal oxide, and as the etch proceeds, the rate changes and deposited oxides etch more slowly. Thus, the researchers propose that a salt layer builds up quickly and remains on the deposited oxide surface, as opposed to thermal oxide with no build-up. The salt is proposed to reduce the flux of the reactants, HF and NH₃, to the surface. After etching, the salt is removed by heating to 100–200°C.

The same reversal in the etching rate is observed for oxides exposed to ion implantation processes [90]. The higher the mass of the ion used, the more damaged the oxide and the lower the amount of oxide etched. The atomic ratio of silicon to fluorine and nitrogen measured by XPS in the (NH₄)₂SiF₆ layer increases with the mass of the implanted ion used. The Si-rich salts are proposed to impede the diffusion of the reactants to the surface.

A dry oxide etching process has been developed using a mixture of NF₃ and NH₃ in a remote microwave plasma to avoid air exposure, which causes oxidation, and avoid damage because of Ar ion bombardment, not caused in the remote plasma. This is performed to replace aqueous HF etching followed by Ar sputter plasma [92]. The process is called SiConi and is used for pre-silicide cleaning to form junctions with low leakage currents. The SiConi process enables the use of materials such as NiSi that are sensitive to the presence of interfacial oxides. Both NF₃ and NH₃ are fed to a remote plasma-generating HF, F atoms, and ammonium fluoride (NH₄F) that react with the SiO₂ film, forming salts such as (NH₄)₂SiF₆. After the etching process, the wafer is heated above 100°C to remove the products from the surface. This process leaves F, which is generated by the low-energy remote plasma, at the interface between NiSi and Si. The F is proposed to limit the formation of the undesired NiSi₂ phase and the shorts that can form between the silicide and the transistor channel, called pipe defects.

6.4 O₃ AND UV/O₃ FOR ORGANIC REMOVAL, RESIST STRIPPING, AND SURFACE OXIDATION

As shown in Table 6.1-1, photochemically enhanced gas-phase reactions find applications in various areas of Si surface cleaning. In many cases, low-pressure mercury (Hg) vapor UV lamps as the radiation sources are sufficient to generate useful chemistries. The general rule is to match the output of the UV source to the energies required for efficient photolysis. The wide range of contaminants that can

be potentially addressed in photochemical cleaning have different excitation energies. Consequently, to allow adequate versatility, the UV sources for photochemical cleaning should ideally feature uniform high-intensity wavelengths ranging from 150 to 600 nm. Therefore, to allow for a wide range of photochemical cleaning applications, UV sources other than low-pressure Hg vapor lamps are needed. However, for any UV source considered, it is important to verify that defects are not generated in the oxides on the surface to be cleaned.

The energy of UV light is close to the strength of a chemical bond; therefore, direct or indirect dissociation can occur when a molecule absorbs light in this energy range. Photolytic processes in the gas-phase and on the surface, a solid phase, are possible. Examples from semiconductor processing include removing carbon-bearing residues and metal atoms using UV/O₃ and UV/Cl₂ and terminating a Si surface with a halogen. The reactors are made from UV-grade fused quartz (e.g., GE, grade 214). An infrared (IR) filter can be used to prevent unintended sample heating, limiting temperature rise to less than 3°C.

An established photochemically enhanced cleaning process is the UV/O₃ removal of organic contaminants. It was first studied as a method to oxidize and volatilize organics from solid surfaces [2,93]. It was later introduced to Si IC processing for a variety of applications, including preoxidation surface treatments [94,95], organic removal before metallization [96], and epitaxial deposition [97]. Overall, the efficacy of this process for surface C and hydrocarbons removal is well documented [98].

Besides effective removal of residual organics, the UV/O₃ process also can be used to strip photoresist and to remove polymeric films remaining on the surface after RIE (reactive ion etching) processes [99]. For these applications, it is essential that 185- and 254-nm wavelength radiation be present in the lamp spectrum to generate atomic O and O₃. These strong oxidizing agents are responsible for both decomposing organic compounds to form CO₂ and causing oxidation of the Si surface. At room temperature, such UV-formed oxide typically does not grow thicker than about 1.5 nm, as shown in Fig. 6.4-22, which makes it useful as a passivating oxide [94,95]. If needed, however, substantially thicker oxides can be grown on Si by UV enhancement at temperatures not exceeding 500°C [100].

6.4.1 History of UV/Ozone Cleaning

In 1972, Bolon and Kunz [1] reported that UV light had the capability to depolymerize a variety of photoresist polymers. The polymer films are enclosed in a quartz tube that is evacuated and then backfilled with O₂. The samples are irradiated with UV light from a medium-pressure Hg lamp that generated O₃. The polymer films of several thousand angstroms thickness are successfully depolymerized in less than 1 h. The major products of depolymerization are found to be H₂O and CO₂. Subsequent to depolymerization, the substrates are

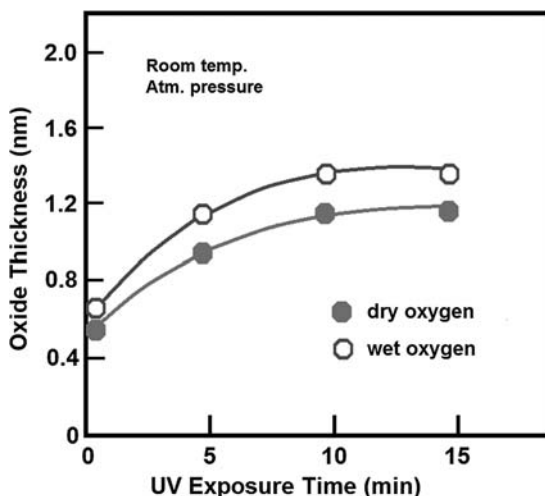


FIGURE 6.4-22 Growth of SiO_2 on the Si surface during exposure to UV radiation in an O_2 -wet and -dry ambient atmosphere with respect to time [94]. Reproduced by permission of ECS—The Electrochemical Society from J. Ruzyllo, G. Duranko, A. Hoff, J. Electrochem. Soc.

examined by AES and are found to be free of carbonaceous residues. Only inorganic residues, such as Sn and Cl, are found. When a Pyrex filter is placed between the UV light and the films or when a N_2 atmosphere is used instead of O_2 , the depolymerization is hindered. Thus, Bolon and Kunz recognized that O_2 and wavelengths shorter than 300 nm played a role in the depolymerization.

During the period 1974–76, Vig et al. [3,4,101] described a series of experiments aimed at determining the optimum conditions for producing clean surfaces by UV irradiation. The variables of cleaning by UV light are defined, and it is shown that, under the proper conditions, UV/O_3 cleaning is capable of producing clean surfaces in less than 1 min. Contact angle measurements, wettability tests, and AES are used to evaluate the results of the cleaning experiments. Most of the experiments are conducted on polished quartz wafers, the cleanliness of which can be evaluated by the “steam test,” a highly sensitive wettability test [101–103]. Contact angle measurements and the steam test can detect fractional monolayers of contamination on a hydrophobic surface. Also tested is a black-light, long-wavelength UV source that only emitted wavelengths above 300 nm. This UV source produced no noticeable cleaning, even after 24 h of irradiation.

In the studies of Vig et al., they found that samples can be cleaned consistently by UV/O_3 only if gross contamination is first removed from the surfaces. The cleanliness of such UV/O_3 -cleaned surfaces has been verified on numerous occasions by AES and XPS [1,3,4,104–106]. Fig. 6.4-23 shows the results from AES before and after UV/O_3 cleaning [104]. Ten minutes of UV/O_3 cleaning reduced the surface contamination on an Al thin film to below the

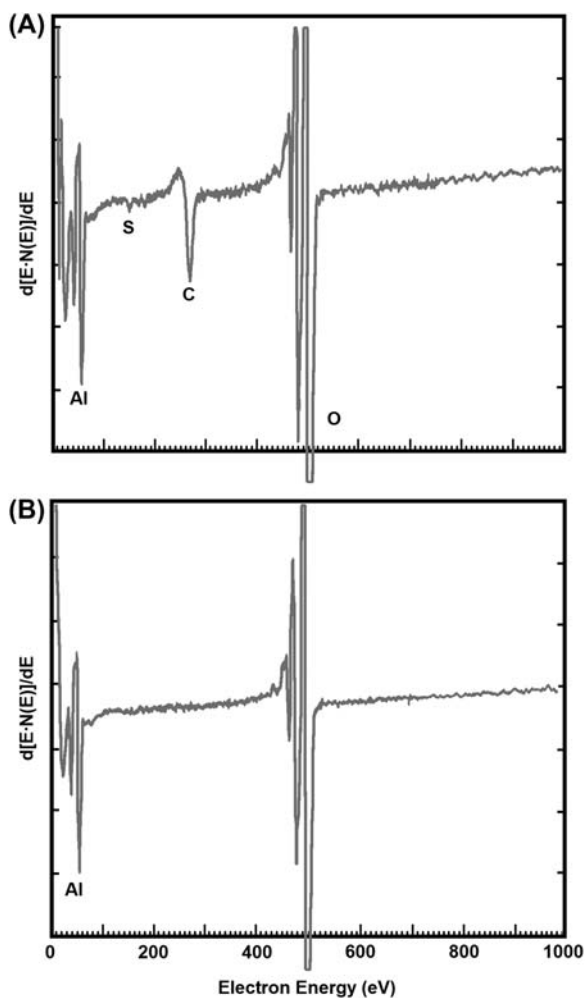


FIGURE 6.4-23 Auger spectra of evaporated Al film on Si surface: (A) before UV/O₃ cleaning and (B) after UV/O₃ cleaning [104]. After cleaning shows the disappearance of the C peak. (Reprinted from R.C. Benson, B.H. Nall, F.G. Satkiewitz, H.K. Charles, *Surface analysis of adsorbed species from epoxy adhesives used in microelectronics*, *J. Appl. Surf. Sci.* 21 219. Copyright (1985), with permission from Elsevier.)

AES detectability level, about 1% of an ML (monolayer). The effectiveness of UV/O₃ cleaning has also been confirmed by ion-scattering spectroscopy/SIMS (ISS/SIMS) [107].

A number of wafers of AT-cut quartz crystal [103] are precleaned and exposed to the UV light until clean surfaces are obtained. Two different UV module configurations are used. Each of these wafers is thoroughly contaminated with human skin oil, which is a difficult contaminant to remove for the IC

TABLE 6.4-5 Exposure Types Versus Cleaning Times to Remove Contamination on the Wafer Surface [108]

Exposure Type	Cleaning Times to Remove Contaminant
“Black light” (<300 nm)	No cleaning
O ₃ , no UV	10 h
253.7 nm, no O ₃	1 h
253.7 nm + O ₃	90 s
253.7 nm + 184.9 nm + O ₃	20 s

Reprinted with permission from Clatterbaugh et al., IEEE Trans. Compon. Hybrids Manuf. Technol. 7 (4) (1984) 349. Copyright 1984, IEEE.

industry. The wafers are precleaned again, then groups of wafers are exposed to each of the four UV/O₃ combinations tested. The time needed to attain a clean surface, as indicated by the steam test, is measured. The samples are placed within 5-mm of the UV source, where the temperature is about 70°C.

In one UV module configuration, the wafers are exposed to 253.7-nm and 184.9-nm light and O₃. In this configuration, the wafers are cleaned in 20 s. In the second UV module configuration, the wafers are exposed to 253.7-nm light and O₃. In the second configuration, the wafers are cleaned in 90 s. Samples exposed to 253.7-nm light without O₃ and to O₃ without UV light are cleaned in 1 and 10 h, respectively. The results are summarized in Table 6.4-5 [108].

Although many organic molecules also absorb radiation at 184.9 nm, it is not possible from these experiments to isolate the cleaning effect of the 184.9 nm radiation. The O₃ concentrations are not measured. The results show, however, that UV/O₃ can be used to remove organic contamination from the substrates in a timeframe that is compatible with semiconductor fabrication.

6.4.2 Wavelengths Emitted by the UV Sources

Because only the light that is absorbed can be effective in producing photochemical changes, the wavelengths emitted by the UV sources are important variables. The low-pressure Hg discharge tubes generate two wavelengths of interest: 184.9 and 253.7 nm. Whether or not these wavelengths are emitted depends upon the lamp envelopes. The emissions through the three main types of envelopes are summarized in Table 6.4-6. Pure quartz is highly transparent to both wavelengths.

The 184.9-nm wavelength is important because it is absorbed by O₂, thus leading to the generation of O₃ [107], and many organic molecules also absorb it. The 253.7-nm radiation is not absorbed by O₂; therefore, it does not

TABLE 6.4-6 Principal Wavelengths of Low-Pressure Hg Discharge Lamps

Wavelength (nm)	Lamp Envelope—Fused Quartz	High-Silica Glass	Glass
184.9	T	O	O
253.7	T	T	O
300.0	T	T	T

O, opaque; T, transparent.

contribute to O₃ generation, but is absorbed by most organic molecules [110,111] and by O₃ [109]. The absorption by O₃ is principally responsible for the destruction of O₃ in the UV chamber. Therefore, when both wavelengths are present, O₃ is continually being formed and destroyed. An intermediate product, both from the formation and the destruction processes, is atomic O, which is a very strong oxidizing agent. The absorption of either or both wavelengths by the organic and other contaminant molecules results in the dissociation or excitation of those molecules. The reaction of the atomic O with excited or dissociated contaminant molecules is believed to be responsible for the cleaning action of UV/O₃, as is discussed below.

The absorption spectrum of O₂ is shown in Fig. 6.4-24 and that of O₃ in Fig. 6.4-25 [109]. The effects of the principal wavelengths generated by low-pressure Hg discharge lamps are summarized in Table 6.4-7. In the studies of Vig et al. [3,4], wafers exposed to 253.7-nm and 184.9-nm light and O₃

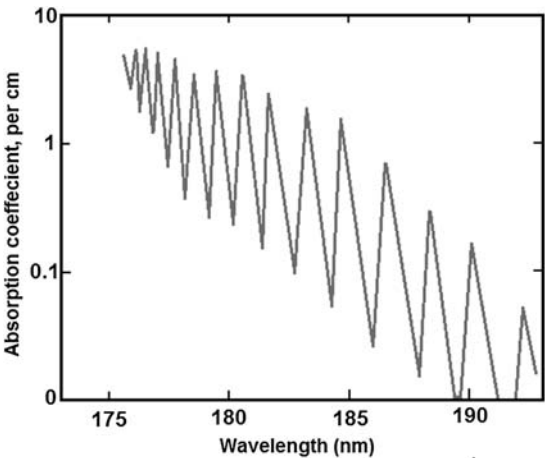


FIGURE 6.4-24 Absorption spectrum of O₂ with respect to UV wavelength [109]. Used with permission from John Wiley and Sons.

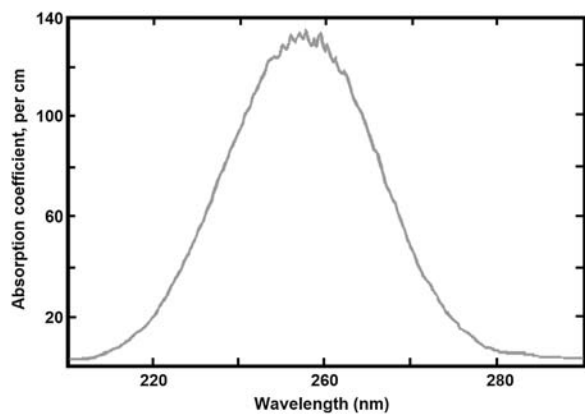


FIGURE 6.4-25 Absorption spectrum of O₃ with respect to UV wavelength [109]. *Used with permission from John Wiley and Sons.*

TABLE 6.4-7 Effects of the Principal Wavelengths Generated by Low-Pressure Hg Discharge Lamp

Wavelength (nm)	Effects
184.9	Absorbed by O ₂ and organic molecules
	Creates atomic O and O ₃
	Breaks contaminant molecule bonds
253.7	Absorbed by organic molecules and O ₃
	Not absorbed by O ₂
	Destroys O ₃
	Breaks contaminant molecule bonds

Used with permission from John Vig.

became clean much faster than the samples exposed to 253.7-nm light and O₃ only, to 253.7-nm light without O₃, or to O₃ without UV light, as summarized in Table 6.4-5. Therefore, although both UV light without O₃ and O₃ without UV light can produce a slow cleaning effect in air, the combination of short-wavelength UV light and O₃, which is obtained from a quartz UV lamp, produces a clean surface orders of magnitude faster.

6.4.3 Distance Between the Sample and UV Source

A variable that can greatly affect the cleaning rate is the distance between the sample and the UV source. In the experiment by Vig et al. [3,4], the UV lamps

are planar sources. Therefore, it can be assumed that the intensity of UV light that reaches a sample is independent of distance. However, this is not so when O_3 is present, because O_3 has a broad absorption band [109,112,113] centered at about 260 nm, as shown in Fig. 6.4-25. At 253.7 nm, the absorption coefficient is 130/cm-atm. The intensity (I) of the 253.7-nm radiation reaching a sample therefore decreases as:

$$I = I_0 e^{-130pd} \quad (6.4-10)$$

where p is the average O_3 pressure between the sample and the UV source at 0°C , and d is the distance to the sample in centimeters. When a quartz UV tube is used, both the O_3 concentration and the UV radiation intensity decrease with increasing distance from the UV source.

Two sets of identically precleaned samples are placed in the second UV configuration described in Section 6.4.1. One set is placed within 5-mm of the UV tube; the other is placed at the bottom of the chamber about 8 cm from the tube. With the O_3 generator off, there is less than a 30% difference in the time for the two sets of samples to attain a minimal (approximately 4 degrees) contact angle; about 60 versus 75 min. When the experiment is repeated with the O_3 generator on, the samples near the tube became clean nearly 10 times faster (about 90 s versus 13 min). Similarly, using the first UV configuration described in Section 6.4.1, samples placed within 5-mm of an O_3 -producing UV tube are cleaned in 20 s versus 20–30 min for samples placed near the bottom of the chamber at a distance of 13 cm. Therefore, to maximize the cleaning rate, the samples should be placed as close as practicable to the UV source.

6.4.4 Organic Contamination Removal

Vig et al. [3,4] tested the effectiveness of the UV/ O_3 cleaning procedure for a variety of contaminants, including residues from cleaning solvents, such as acetone, ethanol, methanol, isopropyl alcohol, trichloroethane, and trichlorotrifluoroethane. After contamination, the wafers are precleaned, then are placed within a few millimeters of the UV tube, and finally exposed to UV/ O_3 . After a 60-s exposure and steam testing, AES indicate that all traces of the contaminants are removed. The precleaning step is used to remove inorganic contaminants, such as dust particles, metals, and salts, that cannot be removed by UV/ O_3 .

For many years, the IC industry has been concerned about C-containing impurities on device wafers. Various methods have been developed for organic impurity removal, most involving UV/ O_3 chemistry [93,97]. As the semiconductor technology evolves, the sensitivity of Si and oxide surfaces to ambient contaminants, especially hydrocarbons, became much more obvious. One such investigation is especially significant because it demonstrated that common laboratory air and materials can produce considerable organic contamination [76]. Furthermore, the investigator proposed that a good

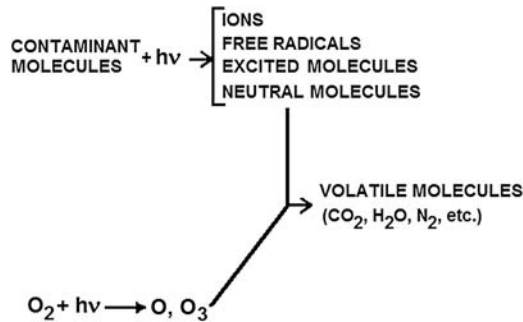


FIGURE 6.4-26 Simplified schematic of the UV O₂ cleaning process. *Used with permission from John Vig.*

percentage of the native oxide regrowth observed using an ellipsometer is not because of oxide growth at all, but merely a build-up of these organics and other impurities.

The use of UV/O₃ chemistries at lower pressure has proven successful in removing unwanted organic impurities from wafer surfaces. Investigations involving these types of precleaning treatments have been carried out in molecular beam epitaxy (MBE) deposition, or conventional low-temperature epitaxial deposition processing [98,114,115]. Some of these investigations have involved activated gas-phase processes, but as noted earlier, plasma and sputter processing tend to cause undesirable effects on device properties.

6.4.5 Mechanism of UV/Ozone Cleaning

The evidence indicates that UV/O₃ cleaning is primarily the result of photosensitized oxidation processes, which is represented schematically in Fig. 6.4-26. The contaminant molecules are excited and/or dissociated by the absorption of short-wavelength UV light. Atomic O and O₃ are produced simultaneously when O₂ is dissociated by the absorption of UV with wavelengths less than 245.4 nm. Atomic O is also produced when O₃ is dissociated by the absorption of the UV and longer wavelengths of radiation [112,113]. The excited contaminant molecules and the free radicals produced by the dissociation of these molecules react with atomic O to form simpler, volatile molecules, such as CO₂, H₂O, and N₂.

The energy required to dissociate an O₂ molecule into two ground state O atoms corresponds to 245.4 nm. However, at and just below 245.4 nm the absorption by O₂ is very weak [109,112,113]. The absorption coefficient increases rapidly below 200 nm with decreasing wavelengths, as shown in Fig. 6.4-24. For producing O₃, a convenient wavelength is 184.9 nm, which is emitted by low-pressure Hg discharge lamps in fused quartz envelopes.

Similarly, most organic molecules have a strong absorption band between 200 and 300 nm; thus, the 253.7-nm wavelength emitted by the same lamp is useful for exciting or dissociating contaminant molecules. The energy required to dissociate O_3 corresponds to 1140 nm; however, the absorption by O_3 is relatively weak above 300 nm. The absorption reaches a maximum near 253.7 nm, shown in Fig. 6.4-25. The actual photochemical processes occurring during UV/ O_3 cleaning are more complex than the schematic shown in Fig. 6.4-26. For example, the rate of production of O_3 by 184.9 nm photons is promoted by the presence of other molecules, such as N_2 and CO_2 . As described previously, the combination of short-wavelength UV light and O_3 produces clean surfaces about 200–2000 times faster than UV light alone or O_3 alone, as shown in Table 6.4-5, and increasing the substrate temperature increases the reaction rates.

The physical and chemical mechanism of hydrocarbon removal by UV/ O_3 cleaning has been studied in an IC processing facility with in situ analysis capabilities [98]. On Si wafers intentionally contaminated with heptanol, volatilization of the hydrocarbons appears to proceed by means of oxidation of the long carbon chain at each carbon atom. In the final stage of the process, the carbon desorbs as CO_2 (but some carbon-containing species remained on the surface). The desorbing gases are found to be primarily CO_2 and H_2O . UV/ O_3 cleaning at elevated ($>60^\circ C$) temperature resulted not only in more complete hydrocarbon removal but also in the removal of surface OH-groups. Mattox [116] found that mild heat increases the UV/ O_3 cleaning rates. However, Bolon and Kunz [1] found that the rate of O_3 depolymerization of photoresists did not change significantly between $100^\circ C$ and $300^\circ C$.

6.4.6 UV/Ozone Cleaning in Vacuum Systems

Sowell et al. [2] reports that, when 10^{-4} Torr pressure of O_2 is present in a vacuum system, short-wavelength UV desorb gases from the walls of the system. During UV irradiation, the partial pressure of O_2 decreases, whereas that of CO_2 and H_2O increases. Similar results are obtained by Hiroki et al. who found that upon switching on a short-wavelength UV lamp in a vacuum chamber, the outgassing of H_2 , CO, CO_2 , CH_4 , etc. increases, whereas H_2O and O_2 slightly reduce [117].

When UV/ O_3 cleaning of Si surfaces in air is compared with UV/ O_3 cleaning in 1–20 Torr of pure O_2 in a vacuum chamber, it is found that, whereas a clean surface can be produced in 7 s in air, it took about 50 min to produce the same cleanliness level in 5 Torr of O_2 . It took 60 min in 20 Torr, and no cleaning effect is observed in 1 Torr after 60 min of cleaning [115]. It should be noted, however, that the cleaning conditions differ; in the air chamber, the UV source is 1-cm from the sample and in the vacuum chamber, the UV source is 6-cm from the sample plus the UV passed through a quartz window before reaching the sample.

A UV/O₃ cleaning method that is suitable for use in an ultrahigh vacuum environment uses a low-pressure Hg UV lamp and a separate O₃ source [118]. The O₃, generated by an O₂ glow discharge at liquid N₂-temperatures, is admitted into the vacuum chamber through a valve. A slow cleaning action is observed at a 2×10^{-5} Torr pressure. Using the same O₃ source, cleaning is also observed without the UV light when the sample is heated to 500 K (227°C). The O₃ molecules that reach the sample decompose on the heated surface. The atomic oxygen created by the decomposition reacts with the contaminant molecules.

Caution should be exercised when using a Hg UV source in a vacuum chamber because, should the lamp envelope break or leak, Hg can enter the chamber, ruining it and creating a hazard because of its toxicity. Mercury has a high vapor pressure; however, its complete removal from a vacuum chamber is a difficult task. Other types of UV sources, such as Xe (xenon) or deuterium lamps, are safer to use in vacuum systems. The UV light can also be radiated into systems through sapphire or quartz windows or through deep-UV fiber optic bundles. A small partial pressure of O₂ should be present during UV cleaning.

Caution must also be exercised when using UV/O₃ in a cryopumped vacuum system because cryopumped O₃ is potentially explosive [119], particularly during regeneration of the cryopump. A convenient method of dealing with this potential hazard is to use two kinds of UV sources, one an O₃-generating source, the other an O₃-destroying source [120].

Integrated processing systems, which incorporate UV/O₃ cleaning, have been proposed [39,98,121–123] and built [124–126] for processing devices in vacuum or in controlled atmospheres.

6.4.7 Surface Oxidation With UV/O₃

Ozone's oxidation power is second only to that of F₂. Ozone can oxidize most inorganic compounds to their highest oxidation state. Extended exposure to UV/O₃ significantly increases the oxide layer thickness on Al surfaces [108]. Whereas the oxide thickness on air-exposed Al surfaces is normally limited to about 50 Å, UV/O₃ exposure increased the oxide layer thickness significantly beyond the normal 50-Å limit, as shown in Table 6.4-8. Passivation of the Al surface may be advantageous.

When the oxidation of Si surfaces is studied by AES, it is found [127] that an etched Si surface can be exposed to air for about 1 h without showing a Si oxide signal at 82 eV. Under UV/O₃ treatment a rapid oxidation takes place. The peak characteristic of Si oxide appears after 1 min of irradiation. The increase of the intensity of this peak gives evidence for a thickening of the layer. Ten minutes of UV/O₃ cleaning increased the oxide thickness on oxidized Si substrates from 0.9 nm (native oxide thickness with no UV/O₃ exposure) to 1.2 nm [97].

TABLE 6.4-8 UV/O₃ Exposure Versus Oxide Thickness on Al [108]

Substrate Treatment	Oxide Thickness (Å)
Evaporate 1 μm of Al (native oxide)	47
10 min UV/O ₃ cleaning	90
60 min UV/O ₃ cleaning	200
Reprinted with permission from Clatterbaugh et al., IEEE Trans. Compon. Hybrids Manuf. Technol. 7 (4) (1984) 349. Copyright 1984, IEEE.	

The O₃ produced by a UV generator is found to enhance the growth rate of SiO₂ during the thermal oxidation of Si at 800°C [128]. After a 140-min oxidation period, the SiO₂ film thickness is 290 Å with O₃ versus 148 Å without O₃. The oxide growth rate enhancement decreases at higher temperatures and longer exposure times (i.e., with thicker films), presumably because during the initial stage of Si oxidation, the reaction at the Si surface may be the controlling factor, whereas in the later stage, the diffusion of reactant through the oxide layer becomes important.

6.4.8 Other Surface/Interface Effects

Energetic radiation such as UV and gamma radiation has been reported to produce dehydration and the formation of free radicals on silica surfaces [129]. However, dehydrated (or siloxinated) silica surfaces are hydrophobic [130,131], whereas UV/O₃-cleaned silica (quartz) surfaces exhibit a very low (<4 degrees) H₂O contact angle, indicating that UV/O₃ does not dehydrate the surfaces, nor does it modify surface silanol groups the way high-temperature vacuum baking does [132]. UV/O₃ has been shown to convert hydrophobic surfaces into hydrophilic surfaces [127].

6.5 UV/CL₂ FOR METALLIC CONTAMINATION REMOVAL

Besides UV/O₃ treatments, there is a growing interest in applying UV stimulation to gas-phase metal removal and native/chemical oxide etching processes. UV exposure of Si in a Cl₂ ambient atmosphere can be used to remove metallic contaminants from the Si surface first by complexing them and then volatilizing the resulting chlorides [42,133,134]. The same complexing and volatilization cycle can be used for Si, as indicated in Fig. 6.5-27, which may help lift metal contaminants off the surface. The role of the lift-off process is confirmed in an investigation in which select trace metals are removed without additional elevated temperature—reduced pressure treatment [135]. However, lift-off is not sufficient to remove alkali metals such as Ca and Na Fig. 6.5-28 demonstrates overall effectiveness of UV/Cl₂

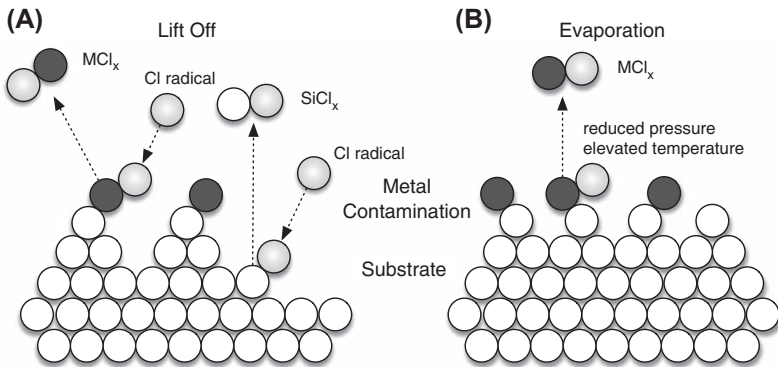


FIGURE 6.5-27 Mechanisms of metal contaminant removal during dry wafer cleaning: (A) formation of metal chlorides and removal through lift-off process and (B) evaporation of metal chlorides at reduced pressure and increased temperature. *Used with permission from Anthony Muscat.*

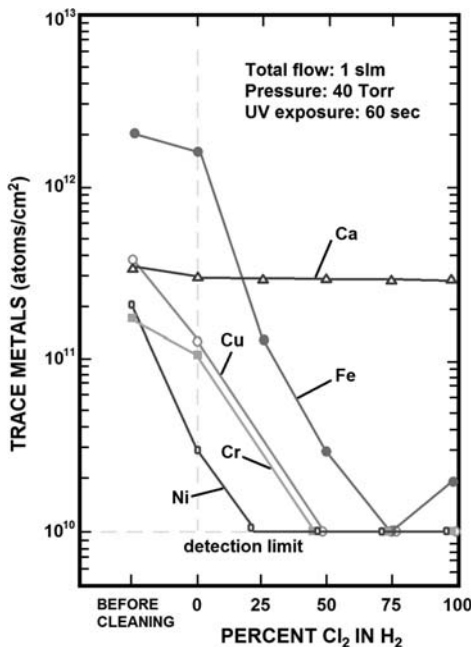


FIGURE 6.5-28 TXRF data showing changes in surface concentration of metallic contaminants on the Si surface as a function of Cl_2 content in H_2 during UV exposure [135]. *Reprinted from Proceedings Microcontamination 92. Copyright 1992, Tom Cheyney, Los Angeles, CA. Used with permission.*

treatment in trace metal removal and at the same time illustrates problems with volatilization of Ca. It is evident that additional treatments following UV/ Cl_2 exposure will be needed to accomplish volatilization of these metals. Alternatively, different chemistries, for instance, based on the use of

organometallic precursors, will have to be developed. These processes must etch Si uniformly and at a low rate to avoid roughening of the surface.

6.5.1 Metallic Contamination Removal

It has long been known that heavy metal impurities result in severe yield and reliability problems in Si devices. These impurities are Fe, Cu, Zn, Ni, Cr, and even Au. A discussion of all metals and their effect on device performance is presented in Chapter 2. They can originate at any process step, including the Si starting material. They will affect a variety of device characteristics, including junction leakage, surface and bulk recombination, emitter to collector shorting, and gate oxide integrity. In addition to heavy metals, light elements, such as C, Al, and S, and alkali metals (Na, K, and Li) can result in catastrophic failures in devices.

It is essential that all metallic impurities be controlled to acceptable minimum concentrations. This control depends on the location of the metal species in or on the device wafer and its chemical nature. For instance, the impurity may be located on top of an oxide layer, within the oxide, at the oxide–Si interface, or within the Si. It also may be in elemental form, or a compound such as an oxide or carbide. Thus, the properties of the metal impurities determine the nature of the optimum removal process, whether it is gettering, liquid-phase, or vapor-phase cleaning. It is possible that cleaning combinations may be appropriate.

Of all the types of impurities, gas-phase removal of metals has proven the most difficult. Part of this difficulty is because of the high temperatures required for vaporization of metallic compounds such as oxides or halides. Also, the location of the metal impurities poses problems. In contrast, aqueous cleaning followed by a H₂O rinse has proven satisfactory because most of the metal reaction products are H₂O-soluble and can be rinsed away. Even so, it is believed that overall possibilities for gas-phase metal removal remain good, especially when considering related particle and organic impurity cleaning processes. A more detailed discussion on mechanisms and future possibilities involving vapor-cleaning techniques follows.

Investigators have reported successful gas-phase removal of metallic impurities from Si wafers. These involve HF chemistries [66,136], combinations of HF and UV/O₃ [137], dry HF–H₂O combinations [81], other gas chemistries [138], and UV-excited Cl₂ processes [139]. Data showing the effectiveness of HF/H₂O cleaning for metal impurity removal from Si surfaces, with and without a post-clean DIW rinse, results as measured by TXRF [81] are shown in Fig. 6.5-29. In the UV/Cl₂ case, photoexcited chlorine species etch a thin Si layer, thus removing metal impurities incorporated on or in the Si. Mechanism of metal removal may be by volatilization of the metal halide or by liftoff. A critical part of this process involves the uniformity and control of etching the Si layer. Data demonstrating this Si etching are shown in Fig. 6.5-30, as reported by Ito [140] using atomic adsorption spectroscopy.

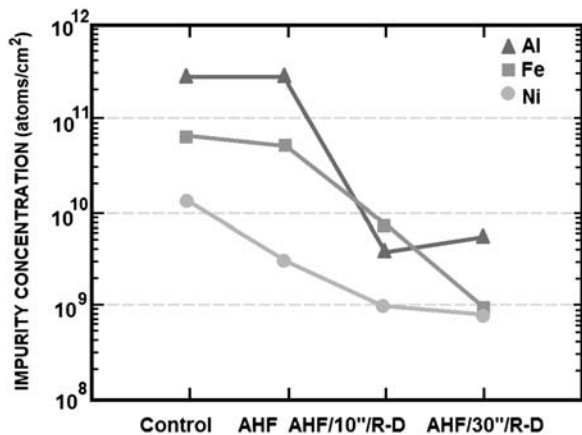


FIGURE 6.5-29 Surface impurity concentrations obtained with anhydrous HF etch and DIW rinse cleaning [81]. AHF is anhydrous HF and R-D is DIW rinse and dry. Copyright 1991 by IEST. Used by permission. Reprinted with permission from 1991 IES Proceedings, published by IEST. www.iest.org (847) 981–0100. From D. Syverson, *An Advanced Dry/Wet Cleaning Process for Silicon Surfaces*, Proceedings IES 37th Annual Technical Meeting, IEST, Arlington Heights, IL, 1991, p. 829.

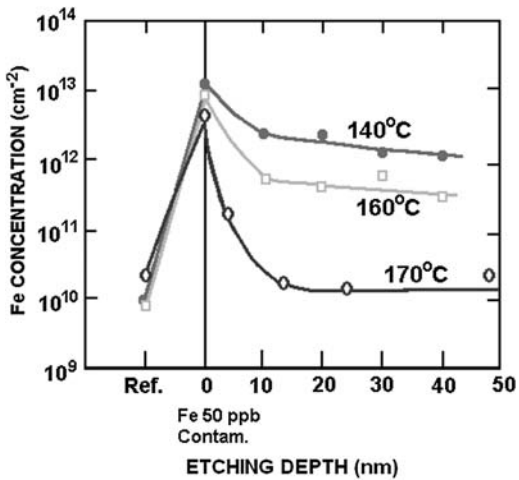


FIGURE 6.5-30 Recovery of cleanliness on Fe-contaminated wafers after photoexcited cleaning [140]. Reference (Ref.) is after a wet cleaning step. The figure shows Fe concentration on a Si surface before and after UV-excited dry cleaning process with and without a rinse. Copyright 1991 by IEST. Used by permission. Reprinted with permission from 1991 IES Proceedings, published by IEST. www.iest.org (847) 981–0100. From T. Ito, *Wafer Dry Cleaning with Photo Excited Halogen Radicals*, Proceedings IES 37th Annual Technical Meeting, IEST, Arlington Heights, IL, 1991, 808.

6.5.2 Mechanisms of UV/Cl₂ Cleaning

Gas-phase metal removal requires elevated temperatures and the formation of a volatile metal compound. For the purposes of this discussion, the assumption is that temperatures above 400°C are unacceptable for typical cleaning processes. Even at this temperature, however, some metals may be volatile enough so that they can simply be thermally desorbed. This is illustrated in Fig. 6.5-31 where the vapor pressures of Kubaschewski and Alcock [141]. The right axis is the equivalent flux, assuming the sticking probabilities are unity for equilibrium vaporization. A vapor pressure of 10^{-6} Torr corresponds to an equivalent flux of approximately 1 ML/s, sufficient for removal of the metal in a reasonable time. From this curve, therefore, all these metals are expected to be volatile enough for a simple thermal cycling to be sufficient to remove them. However, the analysis assumes that the metals are not chemically bonded to Si, O, or other species. This may not be the case for Li, Ca, Mg, and Sr, which form stable oxides and silicides. Na and K oxides are not stable in contact with Si, but the silicides are, which could render this technique ineffective. Of the volatile metals, Zn, Cd, and Pb have the best chance to be successfully thermally desorbed because they do not form stable silicides and their oxides are unstable with respect to Si.

A removal method is the use of Cl₂ (normally excited by UV radiation) to produce volatile species of the metals, which can be desorbed. A plot for selected metal chlorides is shown in Fig. 6.5-32 [139]. The cleaning results

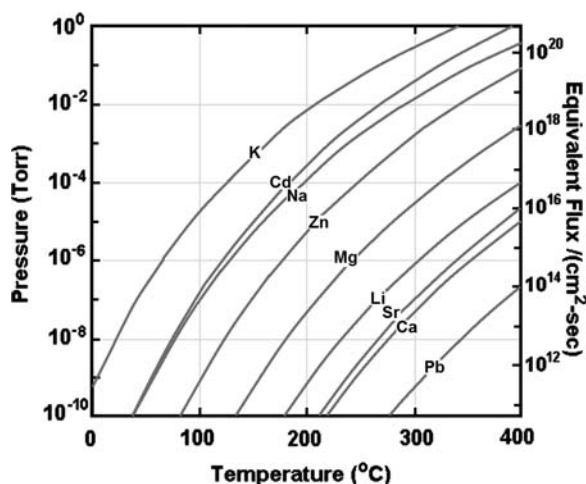


FIGURE 6.5-31 Vapor pressure for “high”—vapor pressure metals as a function of temperature, the right-hand axis corresponds to the equivalent flux leaving the surface assuming a unity equilibrium sticking coefficient [141]. Used with permission from Pergamon Press.

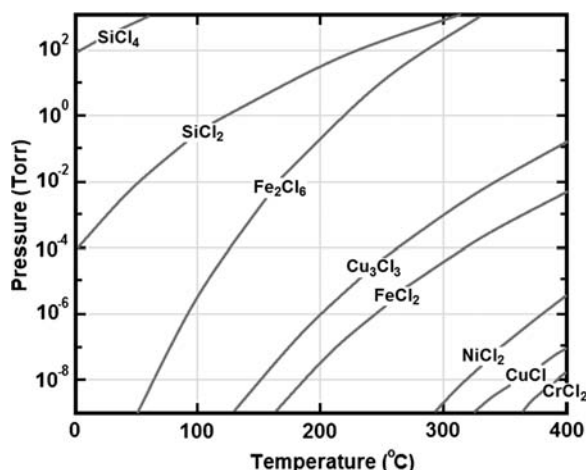


FIGURE 6.5-32 Vapor pressure of some important metal chlorides showing their volatility at temperatures used in UV-enhanced vapor phase metal cleaning [141]. Used with permission from Pergamon Press.

obtained for many of these metals are assumed to be based on evaporation of volatile chlorides. However, significant cleaning efficiencies are observed at 170°C [140], below what is necessary for this mechanism alone (except for possibly Fe). It is likely that the UV radiation or the evaporation of SiCl_x may catalyze the metal chloride removal; it is also possible that some volatile metal–Si–Cl complex may form as well.

One difficulty with this method is related to the rapid reaction and desorption rate of Si in the form of SiCl_2 or SiCl_4 under conditions that lead to efficient metal removal. Fig. 6.5-32 illustrates the SiCl_2 and SiCl_4 vapor pressures, which are higher than those of the metals in the graph at lower temperatures and higher pressure. Metal removal may still be possible without etching the underlying Si if the temperature is kept below about 100°C and the Cl_2 partial pressure below about 1 Torr.

6.6 APPLICATIONS OF GAS-PHASE CLEANING

The initial goal for gas-phase cleaning processes was to replace wet cleans. Since the early stages of development, it became evident that replacing RCA cleans with gas-phase cleans having equivalent particulate, metallic, and organic contamination removal efficiency is an insurmountable task [142]. Gas-phase cleaning processes are advantageous in niche applications where liquid-phase cleaning processes are unable to be used. Some examples of these applications include integrated cluster tool operations; for example, avoiding the collapse of high-aspect ratio structures during drying and other fabrication steps of microelectromechanical systems [143]. Moreover, gas-phase cleaning

is the starting point for the development of atomic layer etch processes that are becoming critical as the scaling of devices pushes the technology to nanoscale dimensions.

Gas-phase cleaning processes offer a new set of possibilities for integration and device engineers to fabricate advanced structures. They provide, in some cases, the initiation path from traditional planar devices to three-dimensional structures, whose formation is impractical using wet processes. Also, gas-phase processes offer solutions to long-known issues of semiconductor fabrication flow, such as divot creation on the edges of shallow-trench-isolation (STI) structures [90].

Gas-phase cleaning processes have found applications in the patterning integration solutions that are being used to keep transistor scaling in accordance to Moore's law [144]. In these applications, gas-phase cleaning processes are typically used to remove the mandrel of the self-aligned multipattern feature, where the integrity of the sidewall spacers that will define the pattern dimension is critical, requiring extreme selectivity without pattern collapse.

6.6.1 Isolation Module Cleaning

The STI structure has been used in CMOS planar technology since the 0.25- μm node. A common issue with STI is “divot” formation at the edge of the isolation oxide. This is a result of more rapid removal of the STI fill oxide compared with the adjacent active area thermally grown oxide when using aqueous HF. Hagimoto et al. [90] showed that using a gas-phase HF and NH_3 process called COR instead of aqueous HF resulted in reduced divot formation. The COR process comprises two steps. The first step uses a gaseous mixture of HF/ NH_3 at controlled temperature and pressure to react with the silicon oxide resulting in the formation of a self-limiting layer of $(\text{NH}_4)_2\text{SiF}_6$. The second step removes the $(\text{NH}_4)_2\text{SiF}_6$ layer by sublimation at temperatures between 100 and 200°C. The COR process has a faster etching rate for thermally grown oxides compared with deposited oxides [90], which is the opposite of what is observed from aqueous HF processes. The reduced removal of the STI fill oxide mitigates the divot formation. In a subsequent study, Saito et al. [91] confirmed that pristine thermally grown oxides etch faster when the COR process is used compared with damaged or defective oxides, again an opposite trend than that obtained if an aqueous HF process is used.

Another application of gas-phase processes is the trimming of the STI liner oxide. In this application the self-limiting characteristic of the COR process allows for a precise and uniform removal of a target amount of the liner oxide that is deposited on the walls of the high-aspect ratio STI trenches before these are filled by deposited oxide. An example of this application is demonstrated by Peng et al. [145] with a target trimming of 50-Å of the oxide liner in trenches with aspect ratio of 6:1 and a depth of 3000 Å for a 28-nm technology.

6.6.2 Gate Module and Prefurnace Cleaning

The purpose of prefurnace (thermal) and pregate cleaning is to remove contaminants that can (1) diffuse into the substrate during the thermal processes, such as metallic contaminants; (2) form compounds that are subsequently difficult to remove, such as SiC; and (3) interfere with the formation of a high-quality grown or deposited gate dielectric film. Generally, it is acceptable to leave a clean oxide on the Si surface before furnace operations, except in the case of epitaxial deposition, which will be discussed in [Section 6.6.3](#).

Integrity of the thin gate oxide is perhaps the most critical property of MOS devices, and it is the most dependent on contaminants such as heavy metals or particles. These contaminants can be present on or in the Si before gate oxidation and can be incorporated into the oxide as it is formed. The oxide integrity normally is reflected by low dielectric breakdown or electrical shorts. Various tests are employed to evaluate gate oxide integrity; these include ramped voltage or field breakdown (V_{BD} or E_{BD}), charge-to-breakdown (Q_{BD}), or time-dependent dielectric breakdown (TDDB).

Based on the device requirements, the prefurnace gas-phase clean must remove metallic and organic contaminants while avoiding excess etching or roughening of the Si surface. Because many contaminants are contained in the chemical or native oxide on the surface of the Si, it is often desirable to treat these incoming surface oxides as contaminants to be removed and replaced with a “clean” oxide during the cleaning process. It should be recognized that dry, gas-phase processes will generally fail to completely remove high levels of contamination and thus will require wet cleaning to remove gross contamination. For instance, it is very difficult to completely remove metal contaminants to a level of 1×10^{12} atoms/cm² with a gas-phase process without significant etching and roughening of the Si surface [146]. Gas-phase cleaning for prefurnace applications should be the final cleaning step performed immediately before thermal treatment to remove any trace contaminants that may have deposited on the surface following the wet cleaning step. Preferably, the wafer is transferred directly from the gas-phase cleaning chamber into the thermal processing chamber without exposure to the ambient environment.

Work by Kasi and Liehr [147] shows that an H-terminated Si surface, prepared by aqueous HF treatment, is highly susceptible to hydrocarbon contamination from the environment. They demonstrated that treatment of the H-terminated surface with UV/O₂ is able to remove the hydrocarbon contamination and prepare the Si surface for gate oxidation. Gate dielectrics grown on the UV/O₂-prepared surface gave equivalent performance to gate dielectrics grown on standard, aqueous RCA-prepared surfaces.

Comparing gas-phase versus conventional aqueous gate oxidation precleans, improved oxide integrity for gas-phase HF precleans has been reported [67,148]. These studies indicated that the gas-phase processes result in a

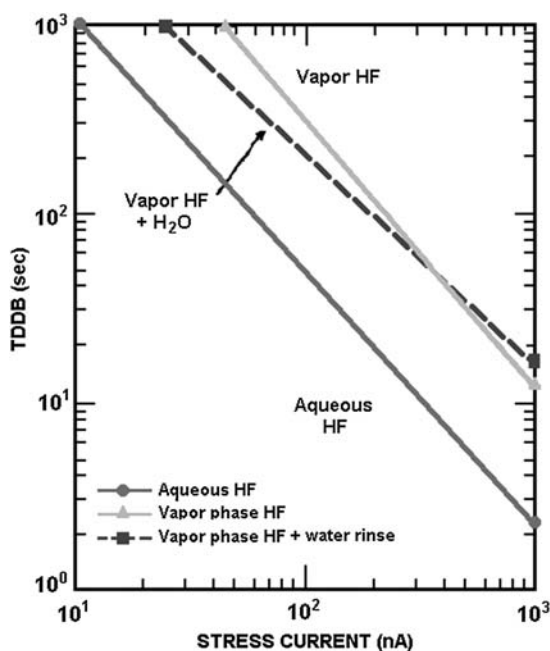


FIGURE 6.6-33 Best fit of present oxidation model to reverse-RCA and HF-cleaned surfaces for 800°C dry O₂ oxidation of Si <100> [49]. Reproduced by permission of ECS—The Electrochemical Society from M. Wong, M.M. Moslehi, D.W. Reed, *J. Electrochem. Soc.*

reduction of particles and other impurities on the surface, which may contribute to this improvement. A typical comparison of TDDB obtained for gas-phase and aqueous precleans is presented in Fig. 6.6-33 [49].

In another study, an atmospheric pressure, gas-phase process using HF, HCl, and H₂O vapor, including spin—rinse—dry, was integrated into a single-wafer gate-stack cluster tool by Frystak and co-workers [149]. After a wafer is precleaned in the atmospheric pressure gas-phase process, it is immediately moved into a vacuum load lock for the gate-stack formation where it receives rapid thermal oxidation for growth of the gate oxide, followed by rapid thermal chemical vapor deposition (CVD) of polySi for the gate electrode. Frystak found that CMOS devices performance, fabricated using this gate stack cluster tool with integrated gas-phase preclean, is equivalent to devices fabricated with conventional batch aqueous cleaning and batch film deposition processes.

A vacuum-based, single-chamber, UV-assisted gas-phase pregate cleaning sequence was developed by Schwab and co-workers for planned integration into a clustered gate stack tool [150]. This sequence started with a short UV/Cl₂ step with control of the wafer temperature to remove surface contaminants. The wafer is then exposed to a mixture of anhydrous HF, IPA (isopropyl alcohol), and H₂O vapor, without UV illumination, to remove native SiO₂ from

the surface. After the surface oxide is removed, the wafer temperature is increased by UV irradiation and then the wafer is exposed to Cl_2 to remove trace metal contaminants. In the final step, a clean surface oxide is regrown by exposure to UV/O_2 . Chamber pressures during these processes are about 100 Torr for the oxide removal and the oxide regrowth steps and about 30 Torr for the metal removal step. Although integration of this process into a gate stack cluster tool is not complete, the results indicate potential as an integrated, all-dry, pregate clean.

It has been noted [14,122,150–152] that Si surface roughness is partially responsible for low dielectric breakdown in gate oxides. This is believed to be because of higher electric fields at nonuniform areas of the oxide, which are caused by the rough Si surface influencing the oxidation uniformity. Surface roughening during gas-phase processing, and especially during UV/Cl_2 steps, has been studied and minimized or eliminated [146].

Another problem for gate oxides is excess carrier trapping in the oxide caused by surface contamination. This is especially critical for memory devices where tunneling oxides are employed. Charge trapping also is also a problem for devices exposed to ionizing radiation. Not only have gas-phase cleaning processes been found to incorporate fewer impurities in the oxides, which can lead to charge or carrier trapping but also F species left on the surface after gas-phase HF cleaning appear to provide trap- or radiation-resistant oxides [153–155]. Somehow, the F species neutralize the oxide traps and significantly retard charge trapping.

Patil et al. [156] have demonstrated that the gas-phase self-limiting HF/NH_3 COR process can be modified to remove the dummy polySi gate that is used on replacement metal gate integration of a 14-nm technology. Although the process details are not disclosed, the researchers present the advantage of gas-phase cleans compared with wet and plasma etch alternatives.

6.6.3 Pre-epitaxial Deposition Cleaning

When several variations of HF treatments are examined for suitability as pretreatments for Si epitaxy processes, the optimum treatment proposed by Suemitsu et al. [157] consists of the sequence of (1) aqueous HF cleaning, (2) DIW rinsing, (3) N_2 gas for drying, and (4) UV/O_3 cleaning. For the production of high-quality epitaxial films by MBE, the cleaning of the Si substrate surfaces is one of the most important steps. UV/O_3 cleaning of Si substrates for Si MBE is found to be effective in producing near-defect-free MBE films [97,115,157–160]. By using UV/O_3 cleaning, the $>1200^\circ\text{C}$ temperatures required for removing surface C by the hydrogen bake method can be lowered to well below 1000°C . The slip lines that result from thermal stresses and the thermal pits that are produced by the high-temperature treatment are minimized at the lower temperature. Impurity redistribution in the substrate is also reduced.

Epitaxial Si deposition following a precleaning step involving gaseous HCl is one of the first examples of integrated processing [25]. Also, MBE scientists used various gas-phase precleans for the preparation of exotic structures—both Si and compound semiconductor-based [161]. Investigators demonstrated the use of in situ gas-phase cleaning processes in combination with lower temperature, about 750°C, for Si epitaxial deposition [161,162]. Because a final H₂O rinse is not necessary, less O, and C and other impurities, are found at the epitaxial Si interface after gas-phase cleaning. The results indicate that integrated processing may be required for contamination-free epitaxial deposits, especially as maximum allowable process temperatures drop even further.

In another study of gas-phase cleaning for epitaxial deposition, UV irradiation using a high-pressure Hg lamp and disilane gas at 20 Torr and 730°C also led to effective surface cleaning before Si epitaxy [163].

6.6.4 Contact/Middle-of-Line Module and Pre-tungsten CVD Cleaning

Gas-phase cleans are used to remove the native oxide before metal deposition for silicide formation. In particular, remaining oxide became a yield factor when nickel silicide was introduced into the CMOS fabrication flow. To remove the native oxide, gas-phase cleans are integrated into the metal deposition cluster platform. The gas-phase cleaning process replaces the sputter etching process that was previously used. An advantage of gas-phase processing is the reduction of the damage to the exposed surfaces because of no bombardment [164].

Interest in gas-phase contact cleaning is driven by the concern that eventually the contact holes will become too narrow and too deep for effective wet-cleaning approaches, although at the 14-nm technology node it appears that wet processes will continue to work for contact cleaning. The feasibility of gas-phase HF/H₂O cleaning following contact dry etching and O₂ plasma stripping is demonstrated by Kwon and Sawin, resulting in good electrical contact to the deposited Al layer [165]. One of the biggest challenges for contact cleaning is minimizing the etching of the dielectric material to avoid enlarging it, while still removing the contamination at the bottom of the contact.

The use of gaseous HF and methanol is proposed as a way to achieve better selectivity between native oxide and the dielectric material forming the contact hole [166]. Methanol, ethanol, and isopropanol are also investigated. In this work, an etching selectivity of BPSG to native oxide of 15:1 is obtained using a gaseous HF/methanol process, while a gaseous HF/H₂O process produces a selectivity of >100:1, while an aqueous buffered HF process produces a selectivity of ~55:1.

After introduction of the gas-phase anhydrous HF plus H₂O vapor process in 1987 [167], for removal of native SiO₂ from polySi surfaces before tungsten

CVD, it became one of the most widespread applications for this technology. Until the introduction of the anhydrous HF preclean process, W CVD was plagued by adhesion problems driven by the inability of aqueous HF and the rinse/dry processes to leave the surface free of residual oxide and watermarks.

Studies have shown that when using an anhydrous HF cleaning step before WSi_x (tungsten silicide) deposition, the amount of O left at the interface under the deposited film is 80% less than the O left after aqueous HF clean followed by a DIW rinse [168]. However, since the introduction of integrated polySi– WSi_x CVD cluster tools, the need for cleaning between the polySi and WSi_x deposition is eliminated by rapid vacuum transfer.

6.6.5 Back-End-of-Line Cleaning

Vias are the openings between each metal interconnect level in an IC device. The requirements for via cleaning steps can depend on what metal layers are being connected because the dielectric film thickness (via depth) and the via diameter may be different for different interconnect levels. Atmospheric pressure, gas-phase HF/ H_2O processes followed by liquid H_2O rinse are used to remove the sidewall residues formed during plasma etching and the Al_2O_3 formed by the plasma stripping process. The devices tested are Al interconnects with TEOS CVD SiO_2 [169]. The goals of the cleaning process are (1) complete removal of polymeric and silicate residues left after dry etching and stripping of the via hole, (2) minimal change in size or shape of the via hole, (3) no damage or degradation of the dielectric film, and (4) low via resistance.

Tong et al. [169] showed that gas-phase HF/ H_2O exposure for less than 5 s followed by a H_2O rinse was able to remove etch residues, as indicated by scanning electron microscope analysis, and also significantly reduce via resistance (40% reduction) and improve via yield compared with a solvent (ethylene glycol + HF) process. The significant reduction in via resistance is believed to be facilitated by the conversion of Al_2O_3 to volatile AlF_3 at the bottom of the via.

Gas-phase HF has not been widely used for via cleaning. Solvent and aqueous-based processes can produce good results for technologies until 14-nm dimensions. Gaseous processes for residue removal after Al etching were being developed at the same time that hydroxylamine-based liquid chemical cleaning solutions were being developed, which proved to be very successful at removing post-etch residue, as discussed in Chapter 10. It was found that gaseous HF mixed with H_2O vapor can be used to remove etch residues and prevent Al corrosion [170]. Although the window for gaseous HF process is somewhat narrower than the hydroxylamine processes, the vapor-phase HF process still found widespread application for manufacturing Al interconnects at the 0.25- μm node. However, its use declined after the development of fluoride-based aqueous cleaning solutions and dilute acid cleaning solutions for Al, discussed in Chapters 4 and 10.

6.6.6 Hard Mask Trim

Hard mask trimming is used in semiconductor fabrication to achieve sublithographic features while maintaining the printed pitch. Its application is mainly for planar CMOS gate definition, where an oxide hard mask is trimmed after being patterned at litho dimensions. Because MOS transistor drive current directly correlates with the gate length, mask trimming for gate definition leads to transistors with shorter gate length and thus higher drive current.

The “trim” of the width of an oxide hard mask that subsequently is used to pattern an MOS gate can be accomplished by a gaseous process using HF and NH₃ to controllably remove thin layers of SiO₂. The trimming process was developed by researchers at IBM and is described in a series of U.S. patents [171–176]. The application of hard mask trimming is described in U.S. patent 6,926,843 and also by Natzle et al. [177]. The gaseous HF/NH₃ exposure is generally carried out at room temperature and at pressures under 15 mTorr. This exposure forms a self-limiting layer of solid reaction product, which is then removed in a heating step at >100°C. The self-limiting room temperature reaction allows controlled thin-film SiO₂ etching. This method is used to create sublithographic features on one part of the device while maintaining the printed dimensions on another part of the chip, which is protected from the gaseous HF/NH₃ exposure.

Hard mask trimming can also be applied to steps other than gate. For instance, it can be used for FinFET fin width definition for advanced nonplanar technologies.

6.6.7 Nanostructure Formation

HF is particularly useful as a highly selective gas-phase etchant for the formation of micro and nanostructures such as accelerometer beams and cylindrical DRAM (dynamic random access memory) capacitors [178,179]. Accelerometers and cylindrical capacitors are often formed with deposited polySi layers on top of sacrificial SiO₂ layers or with sacrificial SiO₂ “molds”. Removal of the sacrificial SiO₂ layer, also known as “release,” leaves these structures freestanding. Etching the sacrificial SiO₂ in liquid etchants is difficult because the surface tension of H₂O causes the released structures to bend, touch, and bond together as H₂O evaporates from the surface [180]. This bending and bonding of small structures is also referred to as “stiction”. Using a gas-phase mixture of HF/H₂O allows the sacrificial oxide to be etched without causing stiction. However, care must be taken in controlling the etching process to ensure that liquid-phase etch byproducts do not form and stiction does not occur.

The vapor from heated aqueous HF, and mixtures of anhydrous HF and H₂O vapor, are used to form accelerometer beams [181–183]. To avoid liquid

formation during the gas-phase etching process, the etching rate and removal of etching products must be carefully controlled by adjusting the reactant concentration and the Si wafer temperature. Because the etching rate must be kept low at about $0.25\text{ }\mu\text{m}/\text{min}$ and the oxide films must be etched laterally on the order of $3\text{--}5\text{ }\mu\text{m}$, the accelerometer release process can take as long as 20 min. Also, as the etching process proceeds the amount of oxide being etched decreases causing a decrease in the amount of gaseous H_2O being produced. Therefore, the ratio of gaseous H_2O to anhydrous HF in the reactant stream must be increased as the etching process proceeds [179].

Depending on the type of SiO_2 used for the cylindrical capacitor mold, it is possible and often desirable, to remove the mold using only anhydrous HF as the reactant. Silicate glass films, such as BSG or BPSG, are hygroscopic and do not require the addition of gaseous H_2O to initiate the etching reaction. Although raising the temperature suppresses the etching of thermal oxide, it is much less effective on suppressing the etching rate of the silicate glass films. Therefore, using anhydrous HF at elevated temperature gives a higher selectivity of silicate glasses compared with thermal SiO_2 [179]. This selectivity is useful where silicate glass is used as the mold layer on top of a SiO_2 layer. Fig. 6.6-34 shows released cylinders approximately 100 nm wide and 1000 nm tall [184].

Controlling etch selectivity of silicate glass over thermal SiO_2 is also useful for trench DRAM capacitor formation. In this case, after the deep trench has been formed by a plasma etching process, the silicate glass hard mask must be removed without lateral etching of the pad oxide layer, which is under the pad

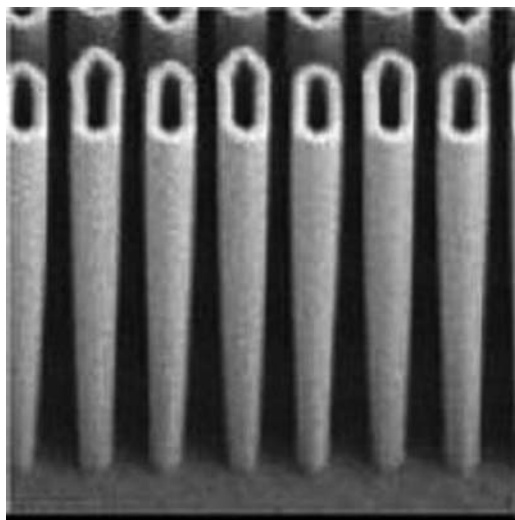


FIGURE 6.6-34 Cylindrical capacitors after removal of the mold layer using gaseous anhydrous HF. *Courtesy of FSI International.*

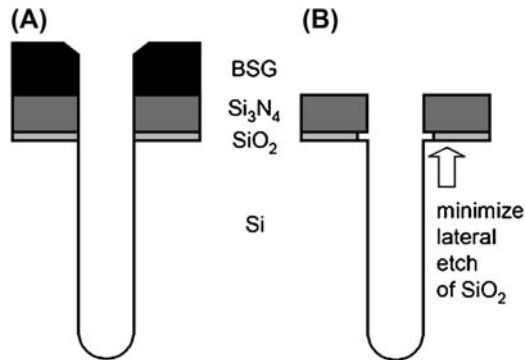


FIGURE 6.6-35 Schematic showing selective hard mask removal for deep-trench capacitor (A) before removal and (B) after hard mask removal [185]. *Courtesy of FSI International.*

nitride and is exposed on the trench sidewall, shown in Fig. 6.6-35. A 60°C, anhydrous HF process is able to remove the BSG hardmask while suppressing lateral etching of the pad oxide [185].

6.6.8 Polysilicon Emitters

The polySi emitter structure used in bipolar device technology has shown great promise but has been difficult to prepare. In this technology, the polySi is deposited onto a single crystal Si region. An n-type dopant, such as arsenic, implanted into the polySi, is diffused into a shallow p-type base region to form the emitter. The presence or lack of a thin oxide layer between the polySi and single-crystal Si regions will affect both current gain and emitter resistance. Results demonstrate that integrated processing permits in situ removal of the initial thin oxide layer by gas-phase HF, regrowing a high-quality 10 Å oxide film, and then depositing the polySi. The resulting bipolar device exhibits tighter distribution of both emitter resistance and current gain [186,187]. Other similar results have been reported [188,189].

6.6.9 Pre-thin Film Measurement

It is shown that some organic contamination can be removed from the surface of Si and SiO₂ by simply heating the wafer to 300°C while being exposed to the radiation from Hg lamps under atmospheric conditions [190–193]. This technique is referred to as rapid optical surface treatment. Although this technique might be used as part of a pregate cleaning sequence, it seems to be most useful as a pretreatment before thin-film measurement [190,193]. Because the thickness of some oxide films used in semiconductor device fabrication has decreased to the level of tens of angstroms, measurement tools have become challenged in accurately and precisely measuring that thickness.

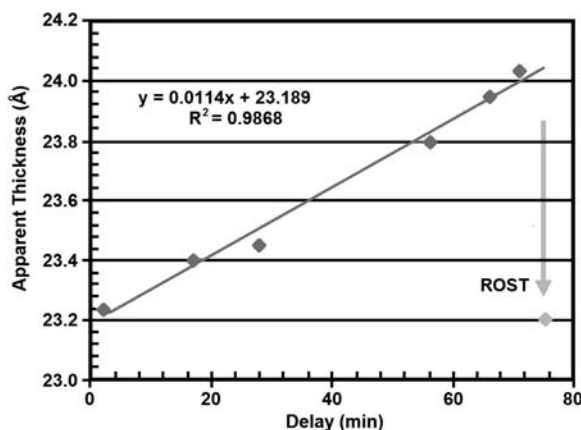


FIGURE 6.6-36 Kinetics of organic contamination deposition on a Si wafer exposed to clean room air, thickness of the contamination layer measured by ellipsometry [191]. A comparison with the rapid optical surface treatment process is shown. *Used with permission from Adrien Danel.*

One problem is interference from the adsorption of a few monolayers of organic contamination from the cleanroom ambient environment. Organic contamination interferes with the ellipsometry reading and can increase the measured thickness by a few angstroms.

Kamieniecki found that exposing the wafer surface to illumination from Hg lamps while heating the wafer to approximately 300°C under atmospheric conditions can remove surface contamination as measured by surface charge analysis [191]. This technique is applied to thin SiO₂ films that are cleaned and then exposed to ambient conditions [190,193]. It is found that in just 60 min ambient exposure, the oxide film thickness can increase by 1 Å, which is a 4% error for a 25-Å-thick film. A 30-s exposure with this technique is able to return the ellipsometry measurement back to that of the freshly cleaned surface, as shown in Fig. 6.6-36 [190].

6.7 PROCESS EQUIPMENT

Integrated processing is the term normally used to describe the combination of two or more sequential processes in semiconductor device fabrication whereby these processes are carried out in situ in a controlled ambient. Often such processes involve a film deposition onto a Si surface, and the resulting interface is critically dependent on the cleanliness of that surface. A typical example might involve a Si cleaning treatment followed by the deposition of a silicide metal such as Ni. A more complex sequence is the preparation of a MOS gate structure, where the integrated processing sequence involves (1) sacrificial oxide strip, (2) gate oxide preclean, (3) gate oxidation, and (4)

polySi gate deposition. Thus, the cleanliness of two critical interfaces ($\text{SiO}_2\text{--Si}$ and polySi--SiO_2) is preserved by controlling the ambient and surface cleanliness throughout the multi-step process sequence.

The cluster tool configuration, used to carry out integrated processing, typically consists of a central automated handling system, which transports device wafers from a loading station to and from various process modules. When the process is complete, the wafers are transported to an unloading station. The wafers are exposed during transport (and in the handler) to only vacuum or an inert gas such as N_2 or Ar. Vacuum level is usually maintained at least to 1×10^{-5} Torr.

Cluster tools are available on various commercially available tools. Several reviews, which discuss the current status and future potential of integrated processing and cluster tool concepts, are available [24,194–204].

Gas-phase equipment has more similarities to plasma etch equipment than to batch wet clean equipment. Single-wafer wet tools incorporate some of these features, opening the possibility of a hybrid tool that combines wet cleaning modules and gas-phase modules.

In an effort to reduce cost, gas-phase equipment aims to eliminate the use of plasma source. This significantly reduces the equipment complexity of gas-phase tools as compared with the platforms that are used for remote plasma etch and chemical downstream etch.

6.7.1 Anhydrous HF Etching Equipment

The industry's first commercial gas-phase processing tool [136,167,205] was designed for the etching of SiO_2 films. The system was introduced commercially in 1987. Operating at ambient temperature and pressure, the single-wafer system provides a controlled mixture of anhydrous HF and H_2O vapor to the process chamber using programmable mass flow controllers. The N_2 carrier gas, anhydrous HF, and the H_2O vapor ratios can be varied to satisfy selectivity requirements for etching dissimilar oxides simultaneously. Fig. 6.7-37 provides a schematic diagram of the key components of the HF vapor etching system. Other systems for this application became available shortly thereafter.

The initial systems used in production were etch-only systems that proved effective for thin-film etching, especially for native oxide removal before CVD processing. An integrated rinse feature, added later, provided improved performance by reducing particles on the wafer and removing metallic and dopant residues resulting from etch processes required for pre-CVD, pregate, and precontact processes [82]. Subsequent product improvements have included the addition of anhydrous HCl for enhanced metallic contamination removal, O_3 gas for light organic cleaning and reoxidation of the Si surface, and a N_2 -ambient wafer staging area and load lock capability for downstream processes.

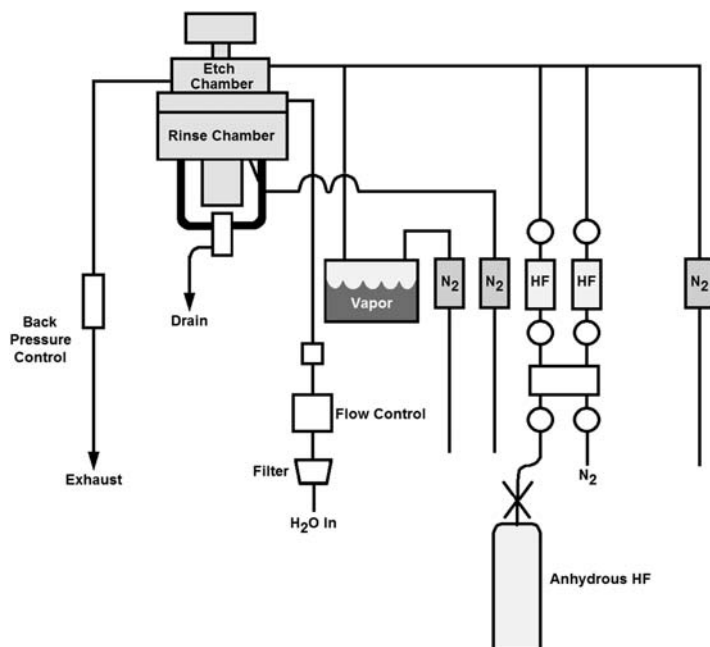


FIGURE 6.7-37 Schematic of an integrated HF vapor system with rinsing capability. *Courtesy of FSI International.*

Applications for the HF vapor etching system have included silicide cleaning, pregate cleans including the etching of the sacrificial oxide before gate oxidation, and premetal contact cleans, especially where CVD technology is chosen over PVD (physical vapor deposition) for first-level metal contacts. Additional applications include pre-epitaxial and pre-polySi deposition cleans and bond pad etching.

Controlled etching rates can vary from 1 to 300 Å/s with total 1σ uniformity of $\pm 2\%$, within wafer, wafer-to-wafer, and batch-to-batch on a 200-Å removal of thermal oxide. The system is particle neutral at sizes $\geq 0.2 \mu\text{m}$. Oxide etching rates as a function of anhydrous HF concentration for various total flow rates are shown in Fig. 6.3-8.

6.7.2 Subatmospheric Gas-Phase HF Etching System

Another type of wafer cleaning system involves gas-phase chemistries based on the vaporization of azeotropic mixtures of HF and/or HCl and H₂O [66]. A schematic of a system for this approach is shown in Fig. 6.7-38. The chamber of this system is about the size of a basketball and is constructed of a specially formulated SiC material. It consists of two hemispheres and a center ring. It

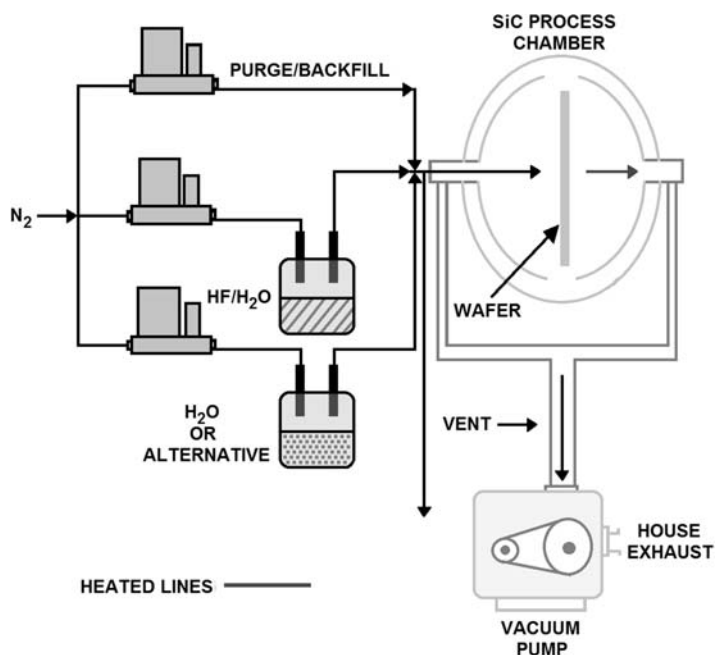


FIGURE 6.7-38 Schematic of a subatmospheric vapor phase wafer cleaning and etching system [66]. Reproduced by permission of ECS—The Electrochemical Society from B.E. Deal, M.A. McNeilly, D.B. Kao, J.M. deLarios, *First International Symposium on Cleaning Technology in Semiconductor Device Manufacturing*.

accommodates one wafer, up to 200 mm in diameter, positioned vertically. Gases are admitted through one side and evacuated from the other. In one version of this system, wafers are maintained at room temperature (22–25°C) but in later versions, higher wafer temperatures are employed. Vaporizers, shown in Fig. 6.7-38, contain solutions of HF/H₂O, HCl/H₂O, H₂O, or solvents such as CH₃OH. The solutions are heated to appropriate temperatures to provide the associated vapor pressures of the reactants. Vapors of these reactants are transported to the chamber by N₂ or Ar, whose flow rates are regulated by mass flow controllers. All materials in contact with the acids are SiC- or Teflon-based. Also shown in Fig. 6.7-38 is a vacuum pump, which permitted the system to be evacuated to the low mTorr-range. Thus, vapor-phase cleaning and etching processes are carried out at reduced pressure, which assisted in vaporization of reactants, reaction products, and contaminants. A typical cleaning sequence started with an initiation step, where the reactants are adsorbed on the Si wafer surface, then the etching initiates; finally, the reactants are desorbed from the surface, leaving a clean surface. In detail, the process proceeds: (1) at the start of the cycle, the wafer automatically is inserted into the

chamber, the door is closed, and the chamber is evacuated to about 1 Torr for a few seconds; (2) reactant gases are then admitted to the chamber to pressures ranging from 100 to 400 Torr for 10–60 s; and (3) the system is then again evacuated to the low mTorr range for a few seconds; (4) after which N_2 or Ar is readmitted to the chamber. The final evacuation or desorption step dries the wafer so that no further drying is necessary. This overall cycle can be varied and is programmed and controlled by a computer.

An improved model of the system employed wafer heating and O_3 activation. The SiC chamber is modified to a single hemisphere with a flat quartz window. This window or plate is covered by a special transparent material that resisted attack of the quartz by HF. The wafer can be exposed to IR or UV for heating up to 400°C. In addition, a corona discharge O_3 source is connected to the inlet lines so that O_3 can be admitted to the chamber before or during the clean–etch treatment cycle. It is found that pretreatment of oxide surfaces using O_3 at elevated temperatures is effective for removing deposited hydrocarbons, also achieving much more uniform etching of Si oxides.

Gas-phase cleaning and etching is shown to eliminate many of the particulate contamination problems, which are associated with liquid-phase cleaning. In general, few, if any, particles are added during gas-phase processes, and often particles are removed. It is also possible to employ in situ monitoring tools, such as residual gas analyzer, ellipsometer, and in-line particle counter. Other experiments involving gas-phase HF processing, along with appropriate equipment, have been reported by van der Heide et al. [206], Onishi, Oki, and co-workers [82,139], Wong et al. [49,148], and by Izumi and co-workers [83,84].

Activated Chlorine: It was mentioned in Section 6.6.3 that vapor HCl has been successfully used for a number of years as a pre-epitaxial Si deposition step for removing surface contamination. It has since been shown that the use of activated Cl_2 gas can be an effective method for removing metal impurities from the Si surface [133,207–211].

Most of the work in this area involves UV activation of Cl_2 gas [133,208–211]. The mechanism involves etching and removing a thin layer of Si, which either reacts or vaporizes the metal atoms as MCl_x along with the Si, or by lifting the metal on top of the evaporating $SiCl_4$. Typically, UV light in the 180- to 50-nm wavelength range is used at temperatures from 100 to 400°C. In Fig. 6.7-39, the relationship of Si removal rate to substrate temperature for Cl_2 with and without UV irradiation is shown [209]. The main challenge in establishing a production-worthy process is the control of the Si etching rate so that a minimum, but uniform layer, is removed.

A second type of Cl_2 -activated etching of Si has been reported using a downstream microwave discharge system [212]. The authors investigated effects of various process variables, such as Si crystal orientations, doping concentration, and substrate temperature. As is the case for UV activation, the main problem for this cleaning process is the ability to uniformly etch Si.

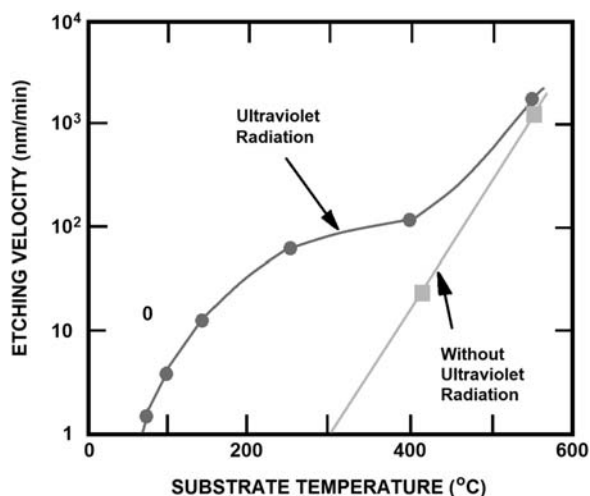


FIGURE 6.7-39 Photochemical etching of Si by Cl_2 gas with and without UV radiation [209]. Graph shows the etching rate (etching velocity) versus temperature. Used with permission from Semiconductor World.

6.7.3 UV/ O_3 Processing Equipment

The materials chosen for the construction of a UV/ O_3 cleaning system should remain corrosion-free after extended exposure to UV/ O_3 . Polished Al with a relatively thick anodized oxide layer, such as Alzak, is one such material. It is resistant to corrosion, has a high thermal conductivity that helps prevent heat buildup, and is also a good reflector of short-wavelength UV. Most other metals, including Ag (silver), are poor reflectors in this range. Oxidation of the materials used for construction is important to avoid because of the highly oxidizing atmosphere. Aluminum oxide coating thicknesses does vary and thicker oxide coatings may stand up to oxidation better than thinner ones. Some commercially available UV/ O_3 cleaners are now constructed of stainless steel. The reflectance of stainless steel in the 200- to 250-nm range is about 20%. To date, no corrosion problems have been reported with such systems. However, UV/ O_3 cleaning systems should be inspected periodically for signs of corrosion to avoid the possibility of particles being generated. Organic materials should not be present in the UV cleaning chamber. Also, the plastic insulation usually found on the leads of UV lamps must be replaced with inorganic insulation such as glass or ceramic.

Low-pressure Hg discharge lamps and an Al stand with Alzak reflectors are examples of commercially available configurations. Lamps should be chosen to produce intensities of short-wavelength UV radiation, about 1.6 mW/cm^2 for a sample position 1-cm from the tube. The module can be injected with O_2

or air; however, it should be completely enclosed to prevent leaking of the O_3 gas and radiation from the light source. The height between the wafer and the UV source are important; an adjustment to bring the wafers close to the UV lamp is an option. Efficient venting and air circulation is important to these modules. The chamber should be enclosed to minimize recontamination by circulating air and to prevent accidental UV exposure and O_3 escape.

The lamp material is important, as it must transmit the wavelength to produce highly reactant O species. Fused quartz transmits both the 253.7- and 184.9-nm wavelengths. In contrast to 184.9 nm, which does not generate measurable O_3 , 253.7 nm does. A typical lamp emits about 0.1 mW/cm^2 of 184.9-nm radiation measured at 1 cm from the tube. A high-silica glass tube made of Corning UV Glass No. 9823 transmits at 253.7 nm but not at 184.9 nm.

The most widely available sources of short-wavelength UV light are Hg arc lamps. Low-pressure Hg lamps in fused pure-quartz envelopes, operating at near room temperature, emit approximately 90% at the 253.7-nm wavelength and generate sufficient O_3 for effective surface cleaning. Approximately 5% of the output of these lamps is at 184.9 nm. Medium- and high-pressure UV lamps [109] generally have a much higher output in the short-wavelength UV range. These lamps also emit a variety of additional wavelengths below 253.7 nm, which may enhance cleaning action. However, they operate at high temperatures (the envelopes are near red-hot), have a shorter lifetime, are of higher cost, and present a greater safety hazard. The Hg tubes can be fabricated in a variety of shapes to fit different applications. In addition to Hg arc lamps, microwave-powered Hg vapor UV lamps are also available [213].

Other available sources of short-wavelength UV include Xe lamps and deuterium lamps. These lamps must also be in an envelope transparent to short-wavelength UV, such as quartz or sapphire, if a separate O_3 generator is not used. In setting up a UV cleaning facility, the UV source should be chosen that generates enough UV/ O_3 to allow for rapid photosensitized oxidation of contaminants. However, too high an output at the O_3 -generating wavelengths can be counterproductive because a high concentration of O_3 can absorb most of the UV light before it reaches the substrates to be cleaned. The substrates should be placed as close to the UV source as possible to maximize the intensity reaching them.

6.7.4 HF/ NH_3 Processing Equipment

The process system for treating a substrate with the HF/ NH_3 COR process incorporates a temperature-controlled chemical treatment chamber with an independently temperature-controlled substrate holder. In this chamber, the substrate is exposed to the gaseous chemistry without plasma, under controlled process parameters that include chamber wall temperature, substrate temperature, chamber pressure, HF flow, and NH_3 flow. The process conditions are

selected to produce the desired reaction between the silicon oxide, HF, and NH_3 resulting in the formation of a self-limited layer of $(\text{NH}_4)_2\text{SiF}_6$. The thickness of the produced layer determines the amount of etched oxide.

The second step of the etch process is desorption of the $(\text{NH}_4)_2\text{SiF}_6$ layer at a temperature $\sim 100^\circ\text{C}$. This can be accomplished in the same chamber that is used to perform the first step by purging all the reaction gases and changing the substrate temperature. Alternatively, the desorption step can be performed in a separate chamber that has controlled ambient pressure and composition and independently tunable substrate and wall temperatures.

Other features that might be present on the processing system are a load lock to load and unload the substrates, a handling system under controlled ambient to prevent particulate contamination during wafer movement, and a cool-down chamber.

Typical systems for HF/ NH_3 processing are described in the patents obtained by Hamelin et al. [214–216].

6.8 SUMMARY

The goal of this chapter is to present an overview of the trends and issues involved in current and future gas-phase wafer cleaning technology. It is important to recognize that this technology will cover only part of the anticipated needs for IC manufacturing. In applications that are less challenging from the point of view of process integration, wet wafer cleaning methods are expected to maintain their present dominant role. Taking into account all of the features outlined in this overview, wafer gas-phase cleaning should be classified as a single wafer process, and, as such, should be considered for practical applications.

One of the important aspects of current and future device fabrication involves the uniform removal or etching of oxide layers. Examples have been presented showing how such etching can be carried out, both in blanket form across a wafer and in selected regions. In addition, this chapter discusses differences in etching rates and reaction mechanisms for different types of silicon oxides and other dielectric films. Moreover, a detailed discussion of mechanisms involved in gas-phase etching, and cleaning in general, has been presented, which should help considerably in the development of more advanced cleaning processes in the future.

The UV/O_3 cleaning procedure has been shown to be a highly effective method of removing a variety of contaminants from Si and compound semiconductor wafers and from many other types of surfaces. It can produce organic-free surfaces at room temperature, either in an ambient pressure or in a controlled atmosphere. The UV/O_3 process is a gas-phase process that creates an oxide similar to the standard RCA wet surface clean. The UV/O_3 process also removes hydrocarbons with similar efficiency as the RCA clean. In combination with a gas-phase method for removing inorganic contamination,

such as cleaning with UV-excited high-purity Cl_2 gas, the method may meet the requirements for the all-dry cleaning methods that will be necessary for processing future generations of semiconductor devices. When compared with plasma cleaning, UV/ O_3 cleaning produces less radiation damage.

In terms of performance, certainly the greatest challenge remaining is the ability of gas-phase cleans to remove all metallic contaminants of concern in Si technology. Although the ability of various gas-phase cleaning processes to remove several contaminant metals has been experimentally confirmed, their effectiveness in volatilization of others elements, such as Na and Ca, still must be unequivocally proven.

Finally, this chapter discusses the importance and requirements for reducing the levels of various types of contaminants using gas-phase cleaning. These impurity types include organics, oxides, particles, and residues and different kinds of metals. It is believed that the difficulty of minimizing these contaminants by vapor cleaning increases in the order listed above. In other words, a form of vapor cleaning, UV light plus O_3 , has been used for some time to remove exposed resist from wafer surfaces. However, at least some of the metals may be difficult to remove completely using known vapor cleaning techniques without subsequent H_2O rinsing.

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